Towards Explainable Ranking with Linear Scores: Efficient Algorithms and Inapproximability

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ABSTRACT

Given a target ranking of items represented by fixed feature vectors, we seek a linear scoring function whose induced order matches the target as closely as possible. We measure disagreement using Kendall's Tau distance, which counts pairs of items whose relative order differs between the two rankings, and also study the complementary objective of maximizing pairwise agreement. Rankings guide decisions in settings such as recruitment, examinations, and sports, where users may want to understand which attributes could account for a given order. Linear scores offer a simple, transparent explanation of a ranking: shared feature weights make each attribute's contribution explicit. Linear and additive explanation models are already well studied, notably through LIME's local models and SHAP's feature attributions. We adopt a single linear score to explain an entire ranking and study the approximation guarantees and computational barriers of finding such a score.

Let n be the number of items and r the dimension spanned by their feature differences. We obtain a deterministic polynomial-time $(r+1)$-approximation for Kendall's Tau disagreement. We also give a randomized $(1-\varepsilon)$-approximation scheme for maximum agreement. For fixed accuracy and success probability, its running time is exponential in r and polynomial in the input size, yielding a polynomial-time approximation scheme when $r = O(\log n)$. We complement these algorithms with hardness results. For some constant $c > 0$, no polynomial-time algorithm approximates either disagreement objective within a factor of $n^{c/\log\log n}$ unless $\mathsf{P} = \mathsf{NP}$. We further establish dimension-dependent lower bounds and rule out deterministic polynomial-time exact algorithms and fully polynomial-time approximation schemes even in logarithmic ambient dimension under the standard parameterized-complexity assumption $\mathsf{FPT} \neq \mathsf{W}[1]$. We complement our theoretical results with extensive experiments with four real-world and seven synthetic datasets.

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1 INTRODUCTION

Ranking is ubiquitous in data systems and determines how data are presented to users. For example, search services order documents, marketplaces order products, recommendation systems order items that compete for a user's attention, and decision-support systems arrange records for inspection. An item's position in such a ranking has practical consequences: higher-ranked items are seen earlier and may be selected for later stages of a workflow. Data-management research has therefore developed efficient methods for ranked retrieval from multiple attribute scores [13, 30, 32, 88].

An important class of ranking models uses *linear scoring functions*. A common weight vector assigns each item a weighted sum of its attributes, and sorting these scores produces the ranking. Linear scores are efficient to evaluate and easy to deploy, while their weights make the contribution of each attribute explicit. They have therefore been widely considered in the literature, including work on indexed linear optimization queries in database systems [13], classical ordinal regression [81], and learning to rank [2, 42, 49].

The ranking presented by a data system, however, need not be generated by a linear score. It may come from a complex nonlinear or black-box model, a legacy or proprietary scoring rule, an editorial process, or a model that uses unavailable features or item-specific adjustments. Even when the resulting target ranking is observable, the mechanism that produced it may be inaccessible, expensive to evaluate, or difficult to understand. It is then natural to seek a simpler model that explains the target ranking by reproducing its order as closely as possible. Linear scores are well suited to this purpose: they are easy to evaluate, and their weights directly describe how the observed attributes influence the ranking. It is therefore natural to ask: *given the items and a target ranking, which linear score induces the closest ranking?*

Ranking explanation and scoring-function synthesis address related questions [35, 89]. In particular, RANKHOW fits a supplied ranking by minimizing positional error through an optimization model and a search over weights [15]. We seek an explanation in terms of the observed attributes by choosing a single weight vector whose induced ranking is as close as possible to the target. Because the target may depend on information or mechanisms not captured by these attributes, exact reproduction may be impossible. We study approximation guarantees and computational limits for finding the closest linear ranking as the number of features grows. Our explanation measures fidelity to the target ranking, rather than recovering the original ranking mechanism.

Example 1.1: Examination ranking

An examination board publishes an overall ranking, but does not reveal its scoring rule. The analyst observes only the mathematics, physics, and chemistry scores, each out of 100. To illustrate one possible hidden rule, suppose that the board first adds the three subject scores and then awards a 15-point additive bonus to candidate B. This bonus depends on information not included in the three observed features.

Candidate	Math.	Phys.	Chem.	Bonus	Target rank	Linear rank
A	90	10	80	0	1	1
B	70	30	60	15	2	3
C	80	20	70	0	3	2
D	60	0	50	0	4	4

Before the bonus, the unit-weight linear score gives totals $180, 160, 170, 110$ for A, B, C, D, respectively. After adding the bonus, the board obtains adjusted scores $180, 175, 170, 110$ and hence the target ranking $A \succ B \succ C \succ D$.
No linear score using only the three subject scores can reproduce this target. Candidate C's feature vector is the average of those of A and B, so every tie-free linear score places C between A and B. In particular, any linear score that places A above B must order them as $A \succ C \succ B$. The unit-weight score produces exactly this ranking and reverses only the target pair B, C. Consequently, it minimizes Kendall's Tau disagreement, with value 1.

In many ranking applications, the central concern is whether pairs of items appear in the correct relative order. *Kendall's Tau distance* captures this notion directly by counting pairs whose order differs between the target and output rankings. It underlies consensus ranking in social choice [44] and has been studied in applications including web search [28], sports, databases, and statistics [1], as well as fair ranking and fair rank aggregation [10, 11, 48, 87]. We therefore focus primarily on this distance in this paper. We also consider *Spearman footrule distance*, the L_1 distance between the position vectors of two rankings. It sums the absolute displacement of every item and therefore reflects the magnitude of positional changes. The distance has been used in database rank aggregation, information retrieval and metasearch, and bioinformatics [5, 31, 66]. It can be computed directly from item positions and is therefore simple and efficient to evaluate. It is well known that Spearman footrule distance lies between Kendall's Tau distance and twice that distance [24]. Thus, Spearman footrule can serve as a simple factor-two proxy for Kendall's Tau when computational convenience is important. In Example 1.1, the Kendall's Tau distance is 1, whereas the footrule distance is 2.

We consider two objectives: *minimum disagreement*, which minimizes the number of target pairs whose relative order is reversed, and *maximum agreement*, which maximizes the number of target pairs whose relative order is preserved. Under Kendall's Tau, the linear ranking in Example 1.1 has one disagreement and five agreements. For exact computation, the two objectives are complementary: on n items, their values sum to $\binom{n}{2}$, so an exact solution to either objective is also exact for the other. However, such a direct relation does not extend to multiplicative approximation, and an approximation for one objective need not approximate the other. For the pairwise-agreement objective, Shin and Todorov give an exact optimization formulation with real-valued weights and binary variables recording pairwise order [77]. Enumerating the orders induced by linear scores also yields an exact $n^{O(d)}$-time algorithm for both objectives, following the region-enumeration approach of Chen et al. [15]; this is polynomial for fixed feature dimension d, but the exponent depends on d, making this approach prohibitive for large d. In this paper, we provide efficient algorithms for arbitrary feature dimension with worst-case approximation guarantees.

1.1 Our Contributions

We give algorithms and lower bounds for both objectives, first for linear scores and then for fixed-degree polynomial scores. Table 1 summarizes our results for linear scores. Our algorithmic guarantees are governed by the *effective dimension*, defined as

$$r := \dim \operatorname{span}\{x_i - x_j : 1 \le i < j \le n\}.$$

This is the number of independent directions among the pairwise feature differences. It satisfies $r \le \min\{d, n-1\}$ and can be much smaller than the number d of supplied features when those features contain redundancies.

- *Approximating minimum disagreement.* We give a deterministic polynomial-time $(r+1)$-approximation for Kendall's Tau distance (Theorem 3.1). The same output is a $2(r+1)$-approximation for Spearman footrule distance (Corollary B.1). The algorithm repeatedly finds a minimal infeasible subset of the desired pairwise comparisons and removes that subset, following the deletion approach for infeasible linear systems [3]. Helly's theorem bounds the number of comparisons removed in one iteration by $r+1$. The sets removed in different iterations are disjoint, and every feasible solution must omit at least one comparison from each; hence their number is at most the optimum.
- *Approximating maximum agreement.* We first obtain a deterministic $1/2$-approximation by comparing the rankings induced by a weight vector and its negation (Theorem 4.1). Our main algorithmic result is a randomized $(1-\varepsilon)$-approximation scheme that succeeds with probability $1-\delta$ and runs in polynomial time when r, ε, and δ are fixed (Theorem 4.4). The scheme samples target comparisons, finds a weight vector that is optimal on the sample, and proves that it performs well on all comparisons. Its running time is FPT in r, $1/\varepsilon$, and $\log(1/\delta)$. When $r = O(\log n)$, it gives a randomized PTAS (Corollary 4.5).
- *Computational limitations.* We complement these algorithms with hardness results showing that significant improvements are unlikely. Unless P = NP, for some constants $c, a > 0$, the minimum-disagreement problem under either distance admits no deterministic polynomial-time $n^{c/\log\log n}$-approximation or $d^{a/\log\log d}$-approximation; the dimension-dependent result holds even when $d = r$. Under certain additional complexity-theoretic conjectures, such as the Projection Games Conjecture [59], it is not possible to attain an approximation factor better than $n^{c'}$ and d^{b} for positive constants c', b (Theorems 5.2 and 5.3). Moreover, even when $d = \Theta(\log n)$, minimizing either distance or maximizing Kendall agreement admits no deterministic polynomial-time exact algorithm or deterministic FPTAS unless FPT = W[1] (Theorem D.2).

Beyond linear explanations: polynomial scores. If a linear score cannot adequately explain the target ranking, it is natural to ask whether a more expressive but still explicit and efficiently evaluable

model can do so. Fixed-degree polynomial scores capture interactions among features and have been considered in information retrieval, quadratic and kernelized learning to rank, and ranking synthesis [15, 34, 43, 71]. The approaches we develop for linear scoring extend to this more general setting. For every fixed maximum degree p, lift each item to its nonconstant monomials of degree at most p, and let r_p be the effective dimension of the lifted items. We obtain deterministic $(r_p + 1)$- and $2(r_p + 1)$-approximations for Kendall's Tau and Spearman footrule disagreement, respectively. The maximum-agreement scheme is FPT in r_p, $1/\varepsilon$, and $\log(1/\delta)$, and becomes a randomized PTAS when $r_p = O(\log n)$. A separate reduction establishes item- and dimension-dependent inapproximability for both minimum-disagreement problems over this entire polynomial scoring family. We provide all the details in Appendix G along with Table 8 summarizing the polynomial-scoring results.

Experimental evaluation. We evaluate our methods on datasets with up to 1.66 million items (Section 6). Our sampling-based variants achieve competitive disagreement against the evaluated baselines; for example, `Sampling` reduces mean disagreement by about 15.5% compared with `RankSVM` on JEE with additive bonuses. To handle larger samples, our practical variants use conflict deletion or local search as alternatives to exact sample optimization. The PTAS guarantee applies to the exact-optimization scheme under the stated sample-size bound.

2 PRELIMINARIES

For a positive integer t, let $[t] = \{1, \dots, t\}$, and let log denote the base-two logarithm.

2.1 Data and linear rankings

Let $X = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^d$ be a set of pairwise distinct feature vectors. A *ranking* is a permutation of the items, or equivalently a strict total order on X. For a ranking ρ, $x \succ_\rho y$ means that ρ places x before y, and $\mathrm{pos}_\rho(x) \in [n]$ denotes the position of x; position 1 is the highest rank.

Our geometric definitions allow arbitrary real coordinates. For bit-complexity bounds, we assume that $X \subseteq \mathbb{Q}^d$, that the coordinates are encoded in binary, and that L_X is the maximum encoding length of one coordinate. In the unit-cost real-RAM model, the factors depending on L_X may be omitted.

The input also contains a *target ranking* π. We number the items in target order, so $x_1 \succ_\pi \cdots \succ_\pi x_n$. Thus, for every index pair $i < j$, the target places x_i before x_j. Let $C = \{(i, j) : 1 \le i < j \le n\}$ be the set of these target comparisons, and let $M = |C| = \binom{n}{2}$. For $e = (i, j) \in C$, define its *comparison vector* by $\Delta_e = \Delta_{ij} = x_i - x_j$.

A *linear scoring function* assigns each item a weighted sum of its features. More specifically, a *weight vector* $w \in \mathbb{R}^d$ assigns score $\langle w, x \rangle$ to item x. Sorting the items by decreasing score produces a ranking. The weight vector agrees with target comparison $e \in C$ exactly when $\langle w, \Delta_e \rangle > 0$. We impose no sign or normalization constraint on w. A weight vector is *tie-free* if it assigns distinct scores to all items. The set of tie-free weight vectors is

$$\mathcal{W}(X) = \{w \in \mathbb{R}^d : \langle w, \Delta_e \rangle \neq 0 \text{ for every } e \in C\}.$$

For $w \in \mathcal{W}(X)$, let σ_w denote its induced ranking. A ranking is *linearly realizable* if a linear score can produce it without ties. More specifically, the family of these rankings is $\mathcal{R}_{\mathrm{lin}}(X) = \{\sigma_w : w \in \mathcal{W}(X)\}$. We assume that the feature vectors are pairwise distinct because identical vectors receive the same score under every linear rule. While our theoretical results assume distinct feature vectors and strict target rankings, our experiments handle ties and duplicates explicitly by excluding pairs tied in the target and counting pairs with identical features as disagreements when the target orders them strictly.

Only score differences determine the order. The *ambient dimension* counts the supplied features, whereas the *effective dimension* counts independent directions among the comparison vectors. Formally, these dimensions are d and $r = \dim V$, respectively, where $V = \mathrm{span}\{\Delta_e : e \in C\}$. Equivalently, r is the dimension of the smallest translated linear space containing X, also called its *affine dimension*. Thus $r \le \min\{d, n - 1\}$. For computation, choose a rational basis matrix P for V and let $y_i \in \mathbb{Q}^r$ satisfy $x_i - x_1 = P y_i$. Then $\langle w, x_i - x_j \rangle = \langle P^\top w, y_i - y_j \rangle$. Because $P^\top$ maps onto $\mathbb{R}^r$, this change of coordinates preserves every realizable ranking. Rational Gaussian elimination computes it in polynomial time.

2.2 Comparison systems and tie-breaking

A *comparison system* requires the weights to put specified pairs in target order, with each score difference at least one. Formally, for $S \subseteq C$, let $\mathcal{L}(S) = \{\langle w, \Delta_e \rangle \ge 1 : e \in S\}$. We call a system *feasible* when one vector satisfies all its inequalities, and *infeasible* otherwise. The right-hand side 1 lets us use standard linear-programming feasibility tests instead of strict inequalities. The following lemma justifies this replacement.

Lemma 2.1. *The strict comparisons $\langle w, \Delta_e \rangle > 0$, $e \in S$, admit a solution if and only if $\mathcal{L}(S)$ has a feasible solution.*

Proof. Any solution of $\mathcal{L}(S)$ satisfies the strict comparisons. For the converse, take a vector satisfying all of them. If S is empty, there is nothing to prove. Otherwise, the finitely many positive numbers $\langle w, \Delta_e \rangle$, $e \in S$, have a positive minimum μ. Replacing w by w/μ raises every required score difference to at least 1. The scaling is positive, so it changes no relative order. □

A feasible comparison system need not decide the order of every pair. The following fact lets both algorithms finish with a strict ranking without losing any comparison they have already secured. It also ensures that tie-breaking does not hide a computational cost.

Lemma 2.2. *There exists a polynomial-time algorithm that, given X, $S \subseteq C$, and a rational solution of $\mathcal{L}(S)$, outputs a rational tie-free weight vector that keeps every comparison in S strictly positive.*

Proof idea. First find a vector v with $\langle v, \Delta_e \rangle \neq 0$ for all $e \in C$. For sufficiently small positive α, the vector $w + \alpha v$ keeps all comparisons in S positive. Each remaining pair forbids at most one value of α, so we can choose a rational value outside these finitely many exceptions. Appendix A.1 gives explicit choices, bit-length bounds, and the polynomial-time implementation. □

2.3 Distances and optimization objectives

For two rankings ρ, σ, the *Kendall's Tau distance* counts pairs that appear in opposite orders. Formally,

$$d_{\mathrm{KT}}(\rho, \sigma) = |\{(x, y) \in X^2 : x \succ_\rho y,\ y \succ_\sigma x\}|.$$

Table 1: Linear-scoring results for Kendall's Tau disagreement and agreement. All approximation algorithms run in polynomial time; the PTAS is randomized. Here $r \le d$ is the effective dimension.

Objective	Setting	Guarantee or lower bound	Condition
Min disagreement	arbitrary r	$(r+1)$-approximation (Theorem 3.1)	–
Max agreement	arbitrary r	1/2-approximation (Theorem 4.1)	–
Max agreement	$r = O(\log n)$	PTAS (Theorem 4.4; Corollary 4.5)	–
Min disagreement	arbitrary dimension	no $n^{c/\log\log n}$- or $d^{a/\log\log d}$-approximation (Theorems 5.2 and 5.3)	$\mathsf{P} \neq \mathsf{NP}$
Min disagreement	arbitrary dimension	no $n^{c'}$- or d^{b}-approximation (Theorems 5.2 and 5.3)	PGC, $\mathsf{P} \neq \mathsf{NP}$
Both objectives	$d = \Theta(\log n)$	no deterministic exact algorithm or FPTAS (Theorem D.2, Appendix D.3)	$\mathsf{FPT} \neq \mathsf{W}[1]$

This formula counts each disagreeing unordered pair once, using its orientation in ρ. The *Kendall agreement* $\mathrm{agr}_{\mathrm{KT}}(\rho,\sigma)$ counts the pairs ordered the same way. Each pair contributes to exactly one count, so

$$\mathrm{agr}_{\mathrm{KT}}(\rho,\sigma) + d_{\mathrm{KT}}(\rho,\sigma) = M. \tag{1}$$

The *Spearman footrule distance* captures the total positional displacements of all items. More specifically,

$$d_{\mathrm{FR}}(\rho,\sigma) = \sum_{x\in X} |\,\mathrm{pos}_\rho(x) - \mathrm{pos}_\sigma(x)|.$$

Diaconis and Graham proved that, for any two rankings ρ and σ [24],

$$d_{\mathrm{KT}}(\rho,\sigma) \le d_{\mathrm{FR}}(\rho,\sigma) \le 2d_{\mathrm{KT}}(\rho,\sigma). \tag{2}$$

We use these bounds to transfer the minimum-disagreement guarantee to footrule distance.

Min-Disagreement Linear Realizability under Kendall's Tau distance asks for a ranking $\sigma \in \mathcal{R}_{\mathrm{lin}}(X)$ minimizing $d_{\mathrm{KT}}(\pi,\sigma)$. We denote its optimal value by

$$\mathrm{OPT}_{\mathrm{KT}}(X,\pi) := \min_{w\in\mathcal{W}(X)} d_{\mathrm{KT}}(\pi,\sigma_w).$$

The Spearman-footrule version is defined analogously by replacing d_{KT} with d_{FR}; we denote its optimal value by $\mathrm{OPT}_{\mathrm{FR}}(X,\pi)$.

Max-Agreement Linear Realizability instead maximizes the number of correctly ordered target pairs. We denote its optimal value by

$$\mathrm{OPT}_{\mathrm{agr}}(X,\pi) := \max_{w\in\mathcal{W}(X)} \mathrm{agr}_{\mathrm{KT}}(\pi,\sigma_w) = M - \mathrm{OPT}_{\mathrm{KT}}(X,\pi).$$

We omit (X,π) from the three optimum notations when the instance is clear. An *α-approximation* returns a solution of cost at most α times the optimum for minimization, and a solution of value at least α times the optimum for maximization. A *polynomial-time approximation scheme* (PTAS) has polynomial running time for each fixed $\varepsilon > 0$ and returns a $(1+\varepsilon)$-approximation for minimization or a $(1-\varepsilon)$-approximation for maximization. If its running time is also a polynomial in $1/\varepsilon$, it is called a *fully polynomial-time approximation scheme* (FPTAS). We call a running time *fixed-parameter tractable* (FPT) in parameters κ when only the multiplicative factor may depend arbitrarily on them. Formally, the running time is $f(\kappa)\,\mathrm{poly}(n,d,L_X)$, with a polynomial exponent independent of κ.

3 APPROXIMATING MINIMUM DISAGREEMENT

In this section, we present a deterministic polynomial-time $(r+1)$-approximation algorithm for minimizing Kendall's Tau disagreement. The difficulty lies in deciding which comparisons to give up: each comparison is easy to impose, but a collection of them may require incompatible weights. Our algorithm repeatedly finds and removes a small infeasible collection of comparisons, following the deletion approach for infeasible linear systems [3]. Its approximation factor depends on the effective dimension rather than on the number of supplied features.

Algorithm 1: Minimal-Conflict(S)

Require: $\mathcal{L}(S)$ is infeasible
$C \leftarrow S$
for *each* $e \in S$ **do**
  if $\mathcal{L}(C \setminus \{e\})$ *is infeasible* **then**
    $C \leftarrow C \setminus \{e\}$
return C

Theorem 3.1. *There exists a deterministic polynomial-time algorithm that, given $X = \{x_1,\dots,x_n\} \subseteq \mathbb{R}^d$ and a target ranking π of X, outputs a linearly realizable ranking $\widehat{\sigma}$. For the effective dimension*

$$r := \dim \mathrm{span}\{x_i - x_j : 1 \le i < j \le n\}$$

of X, the output satisfies $d_{\mathrm{KT}}(\pi,\widehat{\sigma}) \le (r+1)\,\mathrm{OPT}_{\mathrm{KT}}(X,\pi)$.

Note that since $r \le n-1$, the above theorem also implies an n-approximation algorithm.

3.1 Description of the algorithm

A *conflict* is a set $C \subseteq \mathcal{C}$ for which $\mathcal{L}(C)$ is infeasible. We call a conflict *minimal* if $\mathcal{L}(C \setminus \{e\})$ is feasible for every $e \in C$. Thus, no proper subset of a minimal conflict is infeasible; minimality does not mean minimum cardinality among all conflicts.

Algorithm 1 extracts a minimal conflict from any infeasible set S. It starts with all of S, examines the comparisons one at a time, and discards a comparison whenever the remaining set stays infeasible. If removing a comparison makes the system feasible, the subroutine keeps it.

Algorithm 2 repeatedly calls Minimal-Conflict until the remaining target comparisons are jointly feasible. It keeps the still-required comparisons in S and records the discarded comparisons in D. Whenever S contains a conflict, the algorithm removes the whole conflict. It stops once one weight vector satisfies all remaining comparisons, breaks any remaining ties as in Lemma 2.2, and sorts the items by their scores.

Removing a whole conflict may look costly, but the analysis shows that each conflict is small and that the number of iterations is bounded by the optimum.

3.2 Analysis of the algorithm

The analysis first relates the comparisons discarded by the algorithm to the Kendall's Tau objective. It then bounds the number of comparisons discarded in each iteration and the total number of iterations.

Algorithm 2: Closest-Linear-Ranking(X, π)

```
S ← C, D ← ∅
while L(S) is infeasible do
    C ← Minimal-Conflict(S)
    S ← S \ C, D ← D ∪ C
Find a rational solution w of L(S)
Break ties while preserving S
return the ranking obtained by sorting by ⟨w, x_i⟩
```

Consider deleting a set $D \subseteq C$ such that the remaining comparison system $\mathcal{L}(C \setminus D)$ is feasible. We call this task *comparison deletion*, and let $\mathrm{OPT}_{\mathrm{del}}$ denote the minimum possible size of such a set. Deleting a comparison permits the output to disagree with that target pair but does not require a disagreement; hence the final ranking may reverse fewer pairs than the algorithm deletes.

Lemma 3.2 shows that comparison deletion and minimum Kendall's Tau disagreement have the same optimum. Its proof appears in Appendix B.1.

Lemma 3.2. $\mathrm{OPT}_{\mathrm{del}} = \mathrm{OPT}_{\mathrm{KT}}$.

We next bound the cost of one removal. Helly's theorem says that if finitely many convex sets in an r-dimensional space have empty intersection, at most $r + 1$ of them already have empty intersection [55, Section 1.3]. Here a set is *convex* if it contains the line segment between any two of its points. Applied to our comparison systems, Helly's theorem implies that every infeasible system has an incompatible subset of at most $r + 1$ comparisons. Lemma 3.3 turns this geometric fact into the per-iteration bound used in the proof of Theorem 3.1.

Lemma 3.3. *Every minimal conflict C has at most $r+1$ comparisons.*

Proof. Recall that $V = \mathrm{span}\{\Delta_e : e \in C\}$ and $r = \dim V$. The component of w orthogonal to V has zero inner product with every comparison vector, so we may search for w inside V. For each $e \in C$, let $H_e = \{w \in V : \langle w, \Delta_e \rangle \geq 1\}$. Each H_e is convex because a linear inequality defines it. The system is infeasible precisely when $\bigcap_{e \in C} H_e$ is empty.

Apply Helly's theorem in the r-dimensional space V. We obtain a subset $C' \subseteq C$ with at most $r + 1$ elements whose sets already have empty intersection. If C' were a proper subset, C would not be a minimal conflict. Thus $C' = C$, and $|C| \leq r + 1$. □

The size bound applies because Algorithm 1 returns a minimal conflict. Lemma 3.4 proves this property and explains why the algorithm need not revisit a comparison after keeping it.

Lemma 3.4. *Given an infeasible comparison set S, Algorithm 1 outputs a minimal conflict contained in S in polynomial time.*

Proof. The subroutine deletes a comparison only after checking that the remaining system is infeasible. It therefore maintains infeasibility throughout the pass, including at termination.

Now take any comparison e that survives. Let C_e denote the current set when the subroutine examines e. It keeps e because $\mathcal{L}(C_e \setminus \{e\})$ is feasible. Later steps only delete comparisons, so the final set satisfies $C \setminus \{e\} \subseteq C_e \setminus \{e\}$. The same feasible vector satisfies this smaller system. Thus deleting any surviving comparison makes the final system feasible, proving minimality and infeasibility. □

We next bound the number of removals. Every feasible deletion set must contain at least one comparison from each conflict; otherwise it would leave that infeasible collection intact. Our algorithm removes a conflict before finding the next one, so these collections are disjoint. Each iteration therefore forces a different deletion in every optimal solution.

Lemma 3.5. *If Algorithm 2 removes conflicts $C_1, \ldots, C_T$, then $T \leq \mathrm{OPT}_{\mathrm{del}}$.*

Proof. Let $D^\star$ be an optimal deletion set. For any iteration t, suppose $C_t \cap D^\star$ were empty. Then all of C_t would remain in the feasible system $\mathcal{L}(C \setminus D^\star)$. Its feasible vector would also satisfy $\mathcal{L}(C_t)$, contrary to the fact that C_t is a conflict. Hence every intersection $C_t \cap D^\star$ contains at least one comparison.

The algorithm removes all of C_t before the next iteration, so $C_1, \ldots, C_T$ share no comparisons. Choose one element from each $C_t \cap D^\star$. These choices are distinct elements of $D^\star$, which gives $T \leq |D^\star| = \mathrm{OPT}_{\mathrm{del}}$. □

Thus, the algorithm discards at most $r + 1$ comparisons in each iteration and performs at most $\mathrm{OPT}_{\mathrm{KT}}$ iterations. It remains to verify that the final ranking reverses only discarded comparisons and to combine these bounds.

Proof of Theorem 3.1. Let $D = C_1 \cup \cdots \cup C_T$. By Lemma 3.3, $|C_t| \leq r+1$ for every t. Lemma 3.5 gives $T \leq \mathrm{OPT}_{\mathrm{del}}$, and Lemma 3.2 gives $\mathrm{OPT}_{\mathrm{del}} = \mathrm{OPT}_{\mathrm{KT}}$. Therefore,

$$|D| = \sum_{t=1}^{T} |C_t| \leq (r + 1)T \leq (r + 1)\,\mathrm{OPT}_{\mathrm{del}} = (r + 1)\,\mathrm{OPT}_{\mathrm{KT}}\,.$$

When the algorithm stops, its vector satisfies every comparison in $C \setminus D$. Tie-breaking keeps all of them positive. Consequently, every reversed output pair lies in D, and $d_{\mathrm{KT}}(\pi, \widehat{\sigma}) \leq |D|$.

For the running time, each iteration removes a nonempty set, so there are at most M iterations. One call to Minimal-Conflict makes at most M feasibility tests, and the outer loop makes one additional test per iteration. The algorithm therefore uses $O(M^2) = O(n^4)$ linear-programming tests. Each test has at most M rational inequalities in d variables with polynomial encoding length. Polynomial-time linear programming, Lemma 2.2, and score sorting complete the running-time argument. □

As an immediate corollary of the relation between Kendall's Tau and Spearman footrule distances in (2), the same algorithm gives a $2(r + 1)$-approximation for minimum Spearman footrule disagreement; details deferred to Appendix B.2.

4 APPROXIMATING MAXIMUM AGREEMENT

In this section, we present a deterministic 1/2-approximation algorithm and a randomized $(1 - \varepsilon)$-approximation scheme for maximizing Kendall agreement. The approximation scheme samples a small list of target comparisons and finds a weight vector that maximizes agreement on this list. The difficulty is to ensure that the chosen vector also performs well on the full instance. We show that a sample whose size depends on the effective dimension and the accuracy parameters suffices.

For a weight vector $w \in \mathbb{R}^d$, let $F(w) = |\{e \in C : \langle w, \Delta_e \rangle > 0\}|$ denote the number of target comparisons with a positive score difference. This definition allows ties, which do not contribute to the count. For a tie-free vector, $F(w) = \mathrm{agr}_{\mathrm{KT}}(\pi, \sigma_w)$. Recall that $M = |C| = \binom{n}{2}$. We assume $n \geq 2$, since otherwise we can return the unique ranking directly.

4.1 A Deterministic Half-Approximation

We first observe that a linear ranking can always preserve at least half of the target comparisons. Choose any tie-free weight vector and consider the rankings induced by this vector and its negation. Every pair appears in opposite orders in the two rankings, so it agrees with the target in exactly one of them. Returning the better ranking gives the following guarantee.

Theorem 4.1. *There exists a deterministic polynomial-time algorithm that, given $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{R}^d$ and a target ranking π of X, outputs a linearly realizable ranking $\widehat{\sigma}$ satisfying*

$$\mathrm{agr}_{\mathrm{KT}}(\pi, \widehat{\sigma}) \geq \frac{1}{2} \mathrm{OPT}_{\mathrm{agr}}(X, \pi).$$

Moreover, $\mathrm{OPT}_{\mathrm{agr}}(X, \pi) \geq M/2$.

Proof. Apply Lemma 2.2 with $S = \emptyset$ and $w = 0$ to find a tie-free weight vector v. Since negation reverses every pair, $F(v) + F(-v) = M$. Evaluate both rankings and return the one with more agreements. Its agreement is at least $M/2$. Since $\mathrm{OPT}_{\mathrm{agr}} \leq M$, this is at least $\mathrm{OPT}_{\mathrm{agr}}/2$. The returned ranking is linearly realizable, so the optimum itself is at least $M/2$. Tie-breaking, evaluating the agreement counts, and sorting the scores all take polynomial time. □

The lower bound $\mathrm{OPT}_{\mathrm{agr}} \geq M/2$ will also be useful for the approximation scheme. It allows us to convert an additive error proportional to M into a multiplicative error relative to the optimum.

4.2 Uniform Convergence

Let $S = (e_1, \ldots, e_s)$ be a list of comparisons drawn independently and uniformly from C, with replacement. For a weight vector w, let

$$F_S(w) = |\{t \in [s] : \langle w, \Delta_{e_t} \rangle > 0\}|$$

denote its agreement on the sampled list. A comparison may occur several times, and each occurrence contributes separately to $F_S(w)$.

Our algorithm will choose a vector maximizing F_S. To bound its agreement on the full instance, we compare the sampled fraction $F_S(w)/s$ with the full fraction $F(w)/M$. Since the vector is chosen after drawing the sample, this comparison must hold simultaneously for every weight vector. The following lemma bounds the sample size needed for this property.

Lemma 4.2. *There exists a universal constant $c_{\mathrm{samp}} > 0$ such that, for every $0 < \eta, \delta < 1$, a sample of size*

$$s \geq c_{\mathrm{samp}} \frac{r \log(2/\eta) + \log(2/\delta)}{\eta^2} \tag{3}$$

satisfies, with probability at least $1 - \delta$,

$$\left| \frac{F_S(w)}{s} - \frac{F(w)}{M} \right| \leq \eta \quad \text{for every } w \in \mathbb{R}^d. \tag{4}$$

We defer the proof of Lemma 4.2 to Appendix C.1.

Algorithm 3: Approximate-Agreement$(X, \pi, \varepsilon, \delta)$

```
η ← ε/4
Set s to the least integer satisfying (3)
Draw S = (e_1, ..., e_s) independently and uniformly from C, with
  replacement
b ← −1
for each T ⊆ [s] do
    if L({e_t : t ∈ T}) is feasible then
        Find a rational solution w
        Break ties while preserving {e_t : t ∈ T}
        if F_S(w) > b then
            ŵ ← w, b ← F_S(w)
return the full ranking σ_ŵ
```

4.3 The Approximation Scheme

We next describe how to maximize F_S exactly. Enumerate all subsets $T \subseteq [s]$ of sample positions. For each T, test whether $\mathcal{L}(\{e_t : t \in T\})$ is feasible. By Lemma 2.1, this is equivalent to asking whether some weight vector agrees at every position in T. If the system is feasible, find a rational solution and apply Lemma 2.2 to break all ties while preserving these comparisons. Evaluate the resulting vector on sample and retain the candidate with the largest count.

A feasible vector may also agree at positions outside T, so we evaluate $F_S(w)$ rather than use $|T|$ as its value. Repeated comparisons impose the same feasibility constraint, but their occurrences are counted separately when evaluating $F_S(w)$. The following lemma shows that the enumeration finds the sample optimum.

Lemma 4.3. *There exists an algorithm that, given X, π, and a sampled comparison list S of length s, outputs a tie-free rational weight vector $\widehat{w}$ satisfying $F_S(\widehat{w}) = \max_{w \in \mathbb{R}^d} F_S(w)$ and runs in time $2^s \,\mathrm{poly}(n, d, L_X, s)$.*

Proof sketch. Let $w_S^\star$ maximize F_S, and let $T^\star = \{t \in [s] : \langle w_S^\star, \Delta_{e_t} \rangle > 0\}$. Such a maximizer exists because the attainable values of F_S form a nonempty subset of $\{0, \ldots, s\}$. By Lemma 2.1, the comparison system for $T^\star$ is feasible. When the enumeration reaches $T^\star$, its rational solution and subsequent tie-breaking preserve every comparison at these positions. The resulting candidate therefore has sampled agreement at least $|T^\star| = F_S(w_S^\star)$. No candidate can exceed the sample optimum, so retaining the best one gives equality.

There are 2^s subsets, and each iteration uses polynomial-time linear programming, tie-breaking, and evaluation. Appendix C.2 gives the full implementation argument, including repeated comparisons and the encoding length of the returned vector. □

We now combine sampling with this exact optimization procedure. Set $\eta = \varepsilon/4$ and choose s to be the least integer satisfying (3). Draw the sampled list, find a vector maximizing its agreement, and sort all items by the resulting scores. Algorithm 3 records these steps.

The empty subset is always feasible, so the algorithm examines at least one candidate. Tie-breaking makes every candidate a strict ranking of the full input. It remains to show that the best sampled candidate has nearly optimal agreement on all comparisons.

Theorem 4.4. *Let $X = \{x_1, \dots, x_n\} \subseteq \mathbb{Q}^d$, let π be a target ranking of X, and let $r := \dim \operatorname{span}\{x_i - x_j : 1 \le i < j \le n\}$ be the effective dimension of X. There exists a randomized algorithm that, given X, π, and $0 < \varepsilon, \delta < 1$, outputs a linearly realizable ranking $\widehat{\sigma}$, runs in time*

$$2^{O((r\log(2/\varepsilon)+\log(2/\delta))/\varepsilon^2)}\operatorname{poly}(n, d, L_X), \tag{5}$$

and, with probability at least $1-\delta$, satisfies $\operatorname{agr}_{\mathrm{KT}}(\pi, \widehat{\sigma}) \ge (1-\varepsilon)\operatorname{OPT}_{\mathrm{agr}}(X, \pi)$. Algorithm 3 is such an algorithm. Thus, the scheme is FPT jointly in r, $1/\varepsilon$, and $\log(1/\delta)$.

Proof. Let $w^\star$ be an optimal tie-free weight vector for the full instance, and let $\widehat{w}$ be the vector returned by the algorithm. Lemma 4.3 gives $F_S(\widehat{w}) \ge F_S(w^\star)$. Condition on the event (4), which holds with probability at least $1-\delta$. Applying the sampling bound to $\widehat{w}$ and $w^\star$, and using sample optimality, we obtain

$$\frac{F(\widehat{w})}{M} \ge \frac{F_S(\widehat{w})}{s} - \eta \ge \frac{F_S(w^\star)}{s} - \eta \ge \frac{F(w^\star)}{M} - 2\eta.$$

Since $F(w^\star) = \operatorname{OPT}_{\mathrm{agr}}$, this implies $F(\widehat{w}) \ge \operatorname{OPT}_{\mathrm{agr}} - 2\eta M$. By Theorem 4.1, $\operatorname{OPT}_{\mathrm{agr}} \ge M/2$. Together with $\eta = \varepsilon/4$, this gives $2\eta M = \varepsilon M/2 \le \varepsilon \operatorname{OPT}_{\mathrm{agr}}$. Therefore, $F(\widehat{w}) \ge (1-\varepsilon)\operatorname{OPT}_{\mathrm{agr}}$. The returned vector is tie-free, so its agreement count equals $\operatorname{agr}_{\mathrm{KT}}(\pi, \widehat{\sigma})$. The algorithm always returns a linearly realizable ranking; the sampling event is needed only for the approximation guarantee.

For the running time, substituting $\eta = \varepsilon/4$ into (3) gives $s = O((r\log(2/\varepsilon)+\log(2/\delta))/\varepsilon^2)$. By Lemma 4.3, optimizing the sample takes $2^s \operatorname{poly}(n, d, L_X, s)$ time. Computing the effective dimension, drawing the sample, and sorting the final scores take polynomial additional time. Absorbing the polynomial factors in s into the exponential gives (5). The exponent of the remaining polynomial is independent of r, $1/\varepsilon$, and $\log(1/\delta)$, proving the FPT claim. □

4.4 Consequences

For fixed accuracy and failure probability, the exponential factor in (5) depends only on r. When $r = O(\log n)$, this factor becomes polynomial in n, giving the following PTAS.

Corollary 4.5. *Let $X = \{x_1, \dots, x_n\} \subseteq \mathbb{Q}^d$ have effective dimension $r = O(\log n)$, and let π be a target ranking of X. Given $0 < \varepsilon, \delta < 1$, Algorithm 3 returns a linearly realizable ranking $\widehat{\sigma}$ satisfying $\operatorname{agr}_{\mathrm{KT}}(\pi, \widehat{\sigma}) \ge (1-\varepsilon)\operatorname{OPT}_{\mathrm{agr}}(X, \pi)$ with probability at least $1-\delta$, in time $n^{O(\log(2/\varepsilon)/\varepsilon^2)}(2/\delta)^{O(1/\varepsilon^2)}\operatorname{poly}(n, d, L_X)$. For fixed ε and δ, this running time is polynomial.*

Proof. Substitute $r = O(\log n)$ into (5). The factor depending on r becomes $n^{O(\log(2/\varepsilon)/\varepsilon^2)}$, and the remaining exponential factor is $(2/\delta)^{O(1/\varepsilon^2)}$. □

The same output also gives an additive bound on Kendall's Tau disagreement. Recall, agreement and disagreement sum to M. This identity lets us express the guarantee in terms of $\operatorname{OPT}_{\mathrm{KT}}$, although the resulting error depends on the total number of comparisons.

Corollary 4.6. *Given $X = \{x_1, \dots, x_n\} \subseteq \mathbb{Q}^d$, a target ranking π, and $0 < \varepsilon, \delta < 1$, Algorithm 3 returns a linearly realizable ranking $\widehat{\sigma}$ that, with probability at least $1-\delta$, satisfies $d_{\mathrm{KT}}(\pi, \widehat{\sigma}) \le \operatorname{OPT}_{\mathrm{KT}}(X, \pi) + \varepsilon\binom{n}{2}$.*

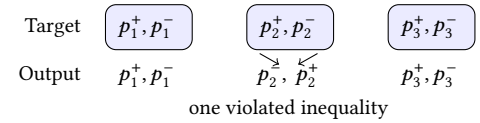


Figure 1: The paired-point construction. If only the second source inequality is violated, a large separating weight keeps the blocks in order and reverses only the second pair: one Kendall's Tau error and two units of footrule distance.

Proof. Subtract the agreement guarantee of Theorem 4.4 from M and use (1). We obtain

$$\begin{aligned} d_{\mathrm{KT}}(\pi, \widehat{\sigma}) \le M - (1-\varepsilon)\operatorname{OPT}_{\mathrm{agr}} &= \operatorname{OPT}_{\mathrm{KT}} + \varepsilon \operatorname{OPT}_{\mathrm{agr}} \\ &\le \operatorname{OPT}_{\mathrm{KT}} + \varepsilon M. \end{aligned}$$

The last inequality uses $\operatorname{OPT}_{\mathrm{agr}} \le M$. □

5 HARDNESS RESULTS

We complement our approximation algorithms with hardness results for minimum disagreement. A reduction from minimizing the number of violated homogeneous strict linear inequalities gives approximation lower bounds for arbitrary dimension. We present the construction and proof sketches below; Appendix D contains the complete proofs. We study exact optimization in logarithmic dimension separately in Appendix D.3.

5.1 Encoding Inequalities as Ranking Errors

We first convert a system of strict linear inequalities into a ranking instance, using two items per inequality. The construction relates the minimum ranking distances to the smallest number of violated inequalities. We also show how to recover weights for the original system from any realizable output ranking. Theorems 5.2 and 5.3 use these properties to transfer approximation lower bounds. Theorem D.2 in Appendix D.3 uses the same construction to prove exact-optimization hardness in logarithmic dimension.

We are given fixed, nonzero rational coefficient vectors $a_1, \dots, a_m \in \mathbb{Q}^q$. We seek a vector $z \in \mathbb{R}^q$ that violates as few inequalities in $\mathcal{I} = (\langle a_\ell, z\rangle > 0 : \ell \in [m])$ as possible. Each inequality is *homogeneous*: its right-hand side is zero, and its left-hand side contains no additive constant. The *violation-count* captures how many inequalities fail, counting repeated occurrences separately. Formally, let $\operatorname{viol}_{\mathcal{I}}(z) := |\{\ell \in [m] : \langle a_\ell, z\rangle \le 0\}|$, and denote its minimum over $z \in \mathbb{R}^q$ by $\operatorname{OPT}_{\mathrm{viol}}$.

Create two points for each inequality: $p_\ell^+ = (m-\ell, a_\ell)$ and $p_\ell^- = (m-\ell, 0)$. Each pair forms a *block*. The target order lists the blocks consecutively, with the positive point first in each: $\pi = (p_1^+, p_1^-, \dots, p_m^+, p_m^-)$. The first coordinate separates the pairs; the remaining coordinates encode the source inequalities. For a weight vector $w = (\lambda, z)$, the score of p_ℓ^+ minus the score of p_ℓ^- is $\langle a_\ell, z\rangle$, since the two items have the same first coordinate. A sufficiently large positive λ keeps all pairs in target block order, as illustrated in Figure 1.

Lemma 5.1 gives exact relations between the optimal ranking values and the minimum violation-count. It also shows how to compute a vector z for the original inequalities from any realizable output ranking, preserving the approximation factor for the minimum-disagreement objective. We need this guarantee because an approximation algorithm may place items from other pairs between p_ℓ^+ and p_ℓ^-, rather than keeping each pair consecutive.

Lemma 5.1. *Given a system $\mathcal{I} = (\langle a_\ell, z\rangle > 0 : \ell \in [m])$ of homogeneous strict inequalities in q variables, the construction above produces $n = 2m$ points in $d = q + 1$ dimensions and a target ranking π that satisfy $\mathrm{OPT}_{\mathrm{KT}} = \mathrm{OPT}_{\mathrm{viol}}$, $\mathrm{OPT}_{\mathrm{FR}} = 2\,\mathrm{OPT}_{\mathrm{viol}}$, and $\mathrm{OPT}_{\mathrm{agr}} = \binom{2m}{2} - \mathrm{OPT}_{\mathrm{viol}}$. There exists a polynomial-time algorithm that, given any realizable strict ranking σ, outputs a rational vector z with $\mathrm{viol}_{\mathcal{I}}(z) \le d_{\mathrm{KT}}(\pi, \sigma)$ and $2\,\mathrm{viol}_{\mathcal{I}}(z) \le d_{\mathrm{FR}}(\pi, \sigma)$.*

We defer the proof to Appendix D.1.

5.2 Inapproximability for Arbitrary Dimension

We show that, when the dimension is part of the input, the minimum-disagreement problems under Kendall's Tau and Spearman footrule distances admit no deterministic polynomial-time $n^{c/\log\log n}$-approximation for some constant $c > 0$, unless P = NP. Lemma 5.1 transfers approximation lower bounds from homogeneous strict inequalities to ranking distances. To apply it, we use the hardness construction for *Learning Halfspaces* (see Appendix D.2), which produces inequality systems whose minimum violation-count is hard to approximate.

Applying Lemma 5.1 yields the following theorem; Appendix D.2 explains how the encoding size determines the exponent constants.

Theorem 5.2. *Consider instances consisting of $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π of X, where d is part of the input. There are constants $c, c' > 0$ such that the minimum-disagreement problems under Kendall's Tau and Spearman footrule distances admit no deterministic polynomial-time $n^{c/\log\log n}$-approximation unless* P = NP. *Under PGC, a deterministic polynomial-time $n^{c'}$-approximation for either objective would also imply* P = NP.

5.3 Dimension-dependent Hardness

We next express the approximation lower bound in terms of the number of features. Using the basis representation from Section 2, we remove redundant coordinates without changing the rankings that a linear score can induce. The resulting instances have ambient dimension equal to effective dimension, so the lower bound holds even when the feature representation has no redundancy.

Theorem 5.3. *Consider instances consisting of $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π of X, and let*

$$r := \dim \mathrm{span}\{x_i - x_j : 1 \le i < j \le n\}$$

be the effective dimension of X. There are constants $a, b > 0$ such that the minimum-disagreement problems under Kendall's Tau and Spearman footrule distances admit no deterministic polynomial-time $d^{a/\log\log d}$-approximation unless P = NP. *Under PGC, a deterministic polynomial-time d^b-approximation for either objective would also imply* P = NP. *These bounds hold even when $d = r$.*

Proof sketch. Construct the paired-point instance from a homogeneous strict inequality system described using S bits. Remove redundant coordinates as in Section 2, and let $d = r$ denote the number of remaining coordinates. This takes polynomial time, preserves all realizable rankings, and leaves both optimum distances unchanged. Moreover, $d = r \le q + 1 \le S + 1$.

An approximation algorithm under Kendall's Tau or Spearman footrule distance on the resulting points therefore returns a ranking realizable on the original points. Lemma 5.1 converts this ranking into weights for the inequalities, preserving the approximation factor for minimizing the violation-count. Choosing $a > 0$ sufficiently small makes $d^{a/\log\log d} < S^{c_0/\log\log S}$ for sufficiently large S. Under PGC, choosing $0 < b < c_1$ similarly gives $d^b < S^{c_1}$. A deterministic polynomial-time algorithm with either guarantee would therefore imply P = NP. We defer details to Appendix D.2. □

Table 2: Dataset description

Dataset	n	d
NBA	22,840	8
Retiring Adult	1,664,500	9
JEE	384,977	3
Human Cognitive Performance	80,000	8
Synthetic datasets	100,000 to 1,000,000	[3, 7, 11, 15]

6 EXPERIMENTS

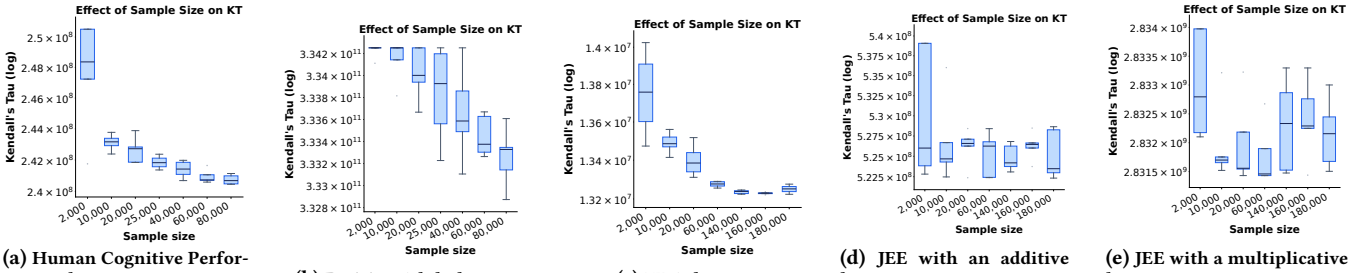
6.1 Experimental Setup

Datasets. We use 4 real-world datasets and 7 synthetic datasets for our experiments. A quick summary of the datasets — their names, sizes, and attributes — is presented in Table 2. Broad descriptions of the datasets are presented below.

- **NBA** [15, 80]: 22,840 player-season tuples (1979/80–2022/23), deduplicated on identical attribute values, ranked on PTS, REB, AST, STL, BLK, FG%, 3P% and FT%. Ground-truth rankings are the MVP ranking and a ranking by MP × PER, which uses attributes outside the ranking set.
- **Retiring Adult** [25]: 1,664,500 U.S. survey records with ten numerically encoded demographic, educational, employment, and financial attributes, ranked by Income in non-increasing order.
- **IIT JEE 2009**: Physics, Mathematics and Chemistry scores of 384,977 candidates, with no official ranking. We rank by the unit-weighted score plus an additive or multiplicative bonus for the minority groups SC and ST [89], chosen by ℓ_2-regularized convex optimization to close the gap to the majority mean, giving two datasets.
- **Human Cognitive Performance** [73]: 80,000 individuals with eight numerical attributes (age, sleep duration, stress level, screen time, exercise frequency, caffeine intake, reaction time, memory test score), ranked by cognitive score in non-increasing order.
- **Synthetic**: Seven datasets with points uniform in a radius-100 sphere, ranked by a linear scoring function and perturbed by a Mallows model [26, 53] with ϕ set to give 5% inversions in expectation. Dimension varies over $\{3, 7, 11, 15\}$ at 10^6 tuples, and size over $\{10^5, 2{\times}10^5, 4{\times}10^5, 10^6\}$ at $d = 3$.

Table 2 captures the dataset size and the number of attributes. A detailed description of our datasets is presented in Appendix H.1.

Baselines. We compare our algorithms against five baselines, all restricted to the same family of linear scoring functions. *Linearized MRC* (LMRC) uses the closed-form, linearized maximum-rank-correlation estimator of Shen et al. [75]: for the target order $x_1 \succ_\pi \cdots \succ_\pi x_n$, it sets $u = \sum_{k=1}^{n}(n + 1 - 2k)x_k$ and returns a ridge-regularized direction $(\widehat{\Sigma} + \lambda I)^{-1}u$, where $\widehat{\Sigma}$ is the empirical feature covariance matrix. *RankSVM* minimizes the regularized pairwise hinge loss [42], whereas *RankNet* uses the pairwise logistic loss [9]. Each method uses all target pairs when practical; otherwise, it optimizes over a uniform sample of pairs, following the scalable



(a) Human Cognitive Performance dataset **(b) Retiring Adult dataset** **(c) NBA dataset** **(d) JEE with an additive bonus** **(e) JEE with a multiplicative bonus**

Figure 2: Varying sample size as we measure Kendall's Tau for different datasets

pairwise learning approach of Sculley [74]. *Kendall local search* directly optimizes the full-data Kendall objective from multiple initial directions, using adaptive coordinate and Gaussian perturbations. It retains the best solution found and accepts non-worsening moves to traverse objective plateaus [57]. Finally, *MaxFS* [16] introduces a nonnegative slack variable for each comparison constraint and minimizes the total slack. It iteratively removes comparisons assigned positive slack until the remaining system is feasible, and then greedily attempts to reinsert the removed comparisons. Every baseline returns a scoring direction and its induced ranking; we recompute disagreement and agreement on the full comparison set. RANKHOW [15] does not scale when the complete ranking list is provided as input, so we do not include it in our experiments.

Baselines for polynomial functions are described in Appendix I.

Details of our algorithms: In this paper, we propose algorithms with varying approximation ratios for KT-error minimization and agreement maximization. While many of these algorithms are theoretically well motivated with guarantees, they are computationally expensive on large datasets. Therefore, we develop the following slight variations of these as the practical counterparts.

`Sampling-CLR`. The theoretically motivated `Closest Linear Ranking` method (Algorithm 2) repeatedly removes Minimal Infeasible Subsystems of Equations/Pairs from the $\binom{n}{2}$ pairs of constraints. The minimal-conflict subroutine removes relatively few constraints per iteration, limiting scalability as n grows. The `Sampling-CLR` variant samples pairs using Equation 3 and applies `Closest Linear Ranking` to the sampled constraints. The resulting linear scoring function is evaluated on the full dataset.

`Sampling`. This variant uses the same random samples as `Sampling-CLR` and searches for a scoring direction using the circle-sweep procedure described below.

Circle-sweep local search. Let $v_k = x_i - x_j \in \mathbb{R}^d$ be the difference vector of the k-th sampled comparison ("i should rank above j") with weight $c_k > 0$, and for a unit vector w let $F(w) = \sum_k c_k \mathbf{1}[w^\top v_k > 0]$ be the weighted sampled agreement. Starting from an initial unit direction w (the weighted centroid of the v_k, its negative, a random direction, or optionally the Linearized MRC direction), each step draws a unit vector $u \perp w$ and considers the great circle $w(\theta) = \cos\theta\, w + \sin\theta\, u$, which lies in the plane $\mathrm{span}\{w, u\} \subset \mathbb{R}^d$. Along it, $w(\theta)^\top v_k = R_k \cos(\theta - \varphi_k)$ with $R_k = \sqrt{(w^\top v_k)^2 + (u^\top v_k)^2}$ and $\varphi_k = \mathrm{atan2}(u^\top v_k, w^\top v_k)$, so comparison k flips exactly at the angles $\varphi_k \pm \pi/2$. Sorting these at most $2m$ angles and sweeping once gives F on every arc of the circle, and we move w to the best arc; since $\theta = 0$ is the current w, F never decreases. After resolving score ties between distinct feature vectors, we compare each candidate with its reverse and return the best ranking, guaranteeing agreement with at least half of the pairs that have distinct features and are strictly ordered by the target.

`Weighted Sampling`. Repeated feature rows can concentrate uniformly sampled item pairs on a small number of feature-vector classes. `Weighted Sampling` groups identical rows into classes and samples comparisons between classes from a mixture of three distributions: the original item-level distribution, a class-balanced distribution, and a distribution favoring small-margin comparisons under a Linearized MRC pilot direction. Each sampled comparison is reweighted by the inverse of its sampling probability, allowing the sampled objective to estimate agreement over the original items. The weighted comparisons are optimized using circle-sweep local search with eight restarts and a maximum of 250 iterations. Candidates and their reverses are then evaluated on the full dataset.

We use the 1/2-approx algorithm as a naive scalable baseline. The `Approximate-Agreement` algorithm with an exact ILP solver approach is used for very small sample sizes of S (like 200).

System details. All our experiments were run on a machine with 256GB RAM and a single AMD EPYC 9654 96-Core Processor, which runs *Ubuntu 24.04.3 LTS*. Our codebase is implemented in `Python`[1].

Experimental questions. We outline the research questions that we answer through our experiments.

(1) **Effect of sample size on quality**: How does sample size affect the quality?
(2) **Scoring models**: Illustrate the effect of different scoring/ranking models like the additive bonus model [89], multiplicative bonus model [89] and Mallows model of perturbation [53] on the quality of the results and time.
(3) **Effect of n and d**: Measure the impact of dataset size and the number of dimensions on quality.
(4) **Scalability**: How do our methods scale as the dataset size grows and the number of dimensions grows?

6.2 Quality-time trade-offs

An important aspect of our sampling-based approach that plays an important role in the quality as well as time is the parameter of the *sample size*. In the first experiment, we show the impact of sample size on KT distance. For this experiment, we vary the sample size starting from 2000 and later in larger increments until timeout in 1800 seconds. KT values are reported only for configurations that complete within the time limit. Each configuration is evaluated using five random seeds: 19, 42, 59, 93, and 123; mean KT values summarize performance across runs.

[1]The codebase can be accessed at https://github.com/surajshetiya/explaining-rankings-linear-and-beyond

Table 3: Per-method Kendall's Tau statistics on the Retiring Adult. Values are reported over 5 runs with different seeds.

Method	Mean ($\times 10^{11}$)	Median ($\times 10^{11}$)	Std. Dev. ($\times 10^{8}$)
Sampling	3.3281	3.3278	2.1689
Weighted Sampling	3.3307	3.3314	2.3413
RankSVM	3.3326	3.3332	1.5452
RankNet	3.3369	3.3373	1.9463
LMRC	3.3425	3.3425	0
Sampling-CLR	3.3626	3.3641	7.3299
1/2-approx	5.7520	5.6525	343.8824

Table 4: Per-method Kendall's Tau statistics on JEE with an additive bonus. Values are reported over 5 runs with different seeds.

Method	Mean	Median	Std. Dev.
Sampling	522652316.4	522597490	426309.3
Weighted Sampling	522678829.8	522371914	648610.2
Sampling-CLR	523907559.8	523865868	1065111.0
RankSVM	618789691.8	611620042	16988646.2
RankNet	767717850.6	764928311	9250349.0
Approx-Agree	789884378.8	795699932	176282141.4
LMRC	4163280176.0	4163280176	0
1/2-approx	10791214208.2	7779312980	8998758783.4

Table 5: Per-method Kendall's Tau statistics on JEE with a multiplicative bonus. Values are reported over 5 runs with different seeds.

Method	Mean	Median	Std. Dev.
Weighted Sampling	2831444352.6	2831430569	19962.6
Sampling-CLR	2831550197.2	2831553181	48458.9
Sampling	2832068468.4	2831608711	786941.2
RankSVM	2833502610.2	2832444548	2687896.5
RankNet	2882495435.4	2875796136	17129872.3
Approx-Agree	3511642947.6	3032369454	733707703.5
LMRC	5853939811	5853939811	0
1/2-approx	12013104560.2	9069898576	8320573155.0

Figure 2 reports the effect of sample size on HCP, Retiring Adult, NBA, and the two JEE rankings. KT error generally decreases as the number of sampled comparisons increases. Both JEE variants exhibit gradual improvements accompanied by substantial variation across runs. Random sampling and local-search initialization may contribute to high variance in JEE settings; however, the experiment does not isolate either effect.

The observed improvements indicate that larger samples can improve the ranking quality (lower KT value) even when the sampled objective is optimized by local search. An interesting observation is that some of our methods are able to process larger sample sizes compared to the others within the 30-minute timeout. We perform systematic experiments later (see § 6.4.2) to determine the relation between d and the sample sizes.

6.3 Effects of different scoring models

We now perform extensive experiments to compare our methods with baselines on different scoring models.

6.3.1 Unknown scoring models. : The HCP and Retiring Adult datasets have real-world features, and the scoring function in both these cases is not known. The NBA dataset has a product-based function which is artificially created in [15] as part of the dataset. Table 3 reports the Retiring Adult results. The HCP and NBA results appear in Appendix H.2.1, in Tables 10 and 9, respectively. Each method is run five times, with mean, median, and standard deviation reported.

On the Retiring Adult dataset (Table 3), MaxFS and LS exceed the time limit. Among completed methods, Sampling achieves the lowest KT error, followed by Weighted Sampling and RankSVM. RankNet, LMRC, and Sampling-CLR obtain slightly higher errors. RankNet and RankSVM exhibit less run-to-run variation than the sampling-based local-search methods. Deterministic LMRC produces the same result across repeated runs but has higher KT error than the leading methods. The 1/2-approximation baseline has the highest error and substantial variability. The full-data Closest Linear Ranking does not scale to any of the datasets due to the large set of pairwise comparisons that need to be processed repeatedly to obtain minimal conflict subsystems. Sampling reduces the problem size for Sampling-CLR, although the conflict-based procedure still processes fewer sampled pairs than the local-search variants within the same budget.

On HCP (Appendix Table H.2.1), MaxFS and Sampling obtain the lowest errors, closely followed by RankSVM and RankNet. Weighted Sampling exhibits the lowest variability among the randomized methods, whereas the 30-minute budget limits Approximate-Agreement. On NBA (Appendix Table H.2.1), LS achieves the lowest error, followed closely by RankSVM, RankNet, and Weighted Sampling. MaxFS exhibits high variability, while LMRC is deterministic but less accurate. Across these real-world datasets, the two sampling-based local-search methods remain competitive, although the best-performing method varies by dataset. The 1/2-approximation baseline consistently has the highest error.

Overall, Sampling and Weighted Sampling perform well overall but are the most consistent among competitors across the different datasets. 1/2 approx performs the worst consistently among the different approaches.

6.3.2 Additive Bonus model. Table 4 shows the results for JEE with an additive bonus. Sampling, Weighted Sampling and Sampling-CLR outperform the competitors by large margins. Other than the deterministic LMRC, Sampling and Weighted Sampling show the most consistency among all randomized methods. Sampling is consistent as it uses local search as a subroutine and waits until convergence. Weighted sampling, due to the optimizations, has an early stopping criterion of 250 iterations with multiple restarts. The deviations can be attributed to the 250 iterations, which could lead to an early stop that is sub-optimal. RankSVM obtains lower error than RankNet but exhibits greater variability. Approximate-Agreement achieves error comparable to RankNet, subject to the sample-size restriction imposed by the time budget. The KT error of LMRC is nearly an order of magnitude larger than that of Sampling. The 1/2-approximation baseline has the highest error and variability.

These results show that our proposed sampling-based variants can recover linear scoring functions with comparatively low disagreement even when the target ranking incorporates group-specific additive adjustments.

6.3.3 Multiplicative Bonus model. The multiplicative model results are presented in Table 5. Weighted sampling and Sampling-CLR closely compete for the top spot, closely followed by Sampling and RankSVM. RankNet does not compete as well as RankSVM. Approximate-Agreement shows a considerably worse KT value

than RankNet. The consistent LMRC has KT values more than twice those of the top competitors. Sampling shows higher deviation than Sampling-CLR and weighted sampling. The local search subroutine could end up in a local minimum and return the local optimum; the subroutine benefits from repeated restarts (8 in the case of weighted sampling). The restarts and 250 iterations help Weighted Sampling to be more consistent.

6.4 Ablation study

For the ablation study, we use the Mallows model. We set the parameter ϕ such that the expected number of inversions is 5%.

6.4.1 Effect of n on quality and time. To measure the effect of n, we maintain the dimension $d = 3$ and vary n over 100,000, 200,000, 400,000 and 1,000,000. We measure the output of the different methods in Table 6. Sampling, Weighted Sampling and LMRC perform the most consistently. As the local search procedure and LMRC sub-routines are based on similar notions, this is likely the reason for these 3 approaches to find the same solution. The solutions found by these 3 methods are comparable to the expected inversion count, showing the effectiveness of these methods. These are closely followed by the Sampling-CLR approach which shows the Greedy solution in Algorithm 2 is equally as effective even with scale and across different datasets. Approximate-Agreement performs considerably worse than Sampling-CLR, but better than 1/2-approx. 1/2-approx outperforms both RankNet and RankSVM under the Mallows model. Among the other competitors, RankNet and RankSVM perform equally well but poorly in comparison to the remaining methods. MaxFS performs well (better than Approximate-Agreement for n=100,000, 200,000 and 400,000), but is not scalable as it does not complete within the 30-minute timeout for the 1 million case.

Among all these methods, Sampling and the weighted version and LMRC perform consistently well under the Mallows model.

6.4.2 Effect of d on quality and time. For this experiment, we keep the number of items in the dataset fixed at 1,000,000 records and vary the number of dimensions — [3, 7, 11, 15]. We record the number of samples that are processed within the 30-minute window. While this is not a monotone function and depends on the sampled pairs, the LP solver's speed and consistency (Sampling-CLR), and other parameters, we consider mean scores over 5 runs to account for some of the variance. The results are presented in Figure 3. Weighted Sampling is the most scalable (around 450,000) at 3 dimensions, but its sample size reduces quickly (to 153,300 at $d = 15$) as the dimension grows. Unweighted Sampling with the Local Search procedure is more stable, and its sample size reduces slowly as the dimension grows. Sampling-CLR has a slightly faster rate of reduction in sample size as dimension grows.

6.5 Proof Of Concept for polynomial functions

Before we conclude our experimental section, we run a small POC to illustrate the correctness of our algorithms. For these experiments, we compute the exact optimal error score and the scores produced by our approaches . We empirically analyze these scenarios to corroborate our theoretical findings (for linear functions). The results for small instances are presented in the Appendix H.3 due to lack of space.

6.5.1 Polynomial functions. Our empirical results for polynomials of degree 2 and 3 are presented in Table 7. MaxFS produces the best result followed by Closest-Poly-Ranking approach for degree 3 polynomial. For degree 2 polynomial MaxFS performs well, followed by Local Search. Sampling follows Local Search for degree-2 polynomial; however, for degree 3 polynomial, the function it does perform even as well as the degree 2 Local Search. This could be attributed to the explosion in dimensions with an increase in dimensions.RankNet, RankSVM, and Smoothed MRC do not perform as well. Modified LMRC performs better than these 3 methods for degree 2, however fares almost as badly for degree 3. Exact ILP-based OPT computation for these experiments is expensive and does not complete even after a one-hour timeout. An interesting observation is that except for Closest-Poly-Ranking and MaxFS other methods behave worse with a 3-degree polynomial than with a 2-degree polynomial.

6.6 Summary of contributions

- Our experiments illustrate the consistency of our algorithms across different datasets. Our methods, like Sampling-CLR, Sampling and Weighted Sampling, consistently produce the top-quality KT results across datasets and scoring models like the Additive bonus model, Multiplicative bonus model and Mallows model.
- Our experiments establish the relationship between sampling size and quality agnostic to the underlying algorithm. This "*empirical*" monotone behavior could guide the user to choose an appropriate sample size.
- Our methods run efficiently for large datasets of 1.5+ million rows, which illustrates the scalability aspect of our algorithms.
- For small settings, we illustrate the effectiveness of our approaches to polynomial models of explanation. Scalability and efficiency of these methods are left as future work.

7 RELATED WORK

Scoring/ranking queries and applications: Numerous models for scoring have been looked at literature. Fagin et al. [32] study the problem of finding top-k efficiently under the class of monotone functions. Skylines [8] and top-k sky-bands [64] capture the structure of the data for efficiently answering top-1 and top-k queries under uncertain user preferences. Chang et al. [13] examine efficient methods for answering linear optimization queries using repeated hull peeling and indexing techniques.

Problems in multi-criteria decision-making over linear functions, such as adding fairness to queries [6], finding data representatives [7, 62, 76], reverse top-k queries [38, 52], Why-Not queries [14, 36, 50, 51], Provenence queries [23], etc., have been extensively studied. Faulkner et al. [33] generalize finding data representatives to the class of quadratic and non-linear functions. Peng et al. [65] have extensively studied top-k queries over probabilistic linear and non-linear queries. With a large body of work on linear and polynomial function classes, we use linear/polynomial functions as the basis for explainability.

Learning to rank: Ordinal regression fits ordered responses through linear programming [81], while RankSVM fits pairwise preferences using a support-vector objective [42], with scalable linear implementations [2, 49]. The support-vector methods combine

Table 6: Mean and standard deviation of KT inversion counts across uniform synthetic datasets when varying n. The final row gives the 5% inversion reference as an expected value of the true KT score.

Method	$n = 100,000$ $\times 10^5$	$n = 200,000$ $\times 10^5$	$n = 400,000$ $\times 10^6$	$n = 1,000,000$ $\times 10^7$
Sampling	2492.1 ± 4.2	10000.9 ± 4.7	3999.6 ± 2.5	2499.9 ± 3.7
Weighted Sampling	2492.1 ± 4.2	10000.9 ± 4.7	3999.6 ± 2.5	2499.9 ± 3.7
LMRC	2492.1 ± 4.2	10000.9 ± 4.7	3999.6 ± 2.5	2499.9 ± 3.7
Sampling-CLR	2492.9 ± 4.6	10002.1 ± 4.9	4000.5 ± 1.9	2500.5 ± 3.1
MaxFS	2494.8 ± 4.0	10016.9 ± 4.6	4005.0 ± 3.6	–
Approx-Agree	2833.9 ± 163.9	11082.8 ± 403.0	4507.5 ± 299.5	2733.0 ± 34.4
1/2-approx	18977.5 ± 1245.4	60139.7 ± 343.8	21395.2 ± 3389.7	14006.3 ± 3417.9
RankNet	24399.4 ± 1917.0	82878.0 ± 2679.2	34382.8 ± 7412.6	19105.4 ± 10629.1
RankSVM	24399.4 ± 1917.0	82878.0 ± 2679.2	34382.8 ± 7412.6	19105.4 ± 10629.1
Expected KT value (5% inversion)	2499.975	9999.95	3999.99	2499.9975

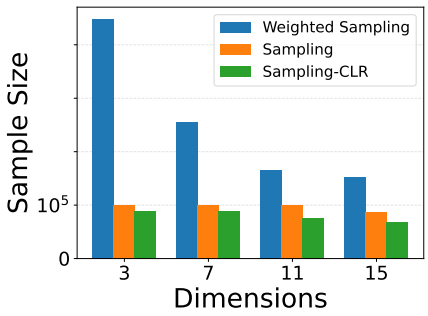


Figure 3: Sample size when varying dimensions

Table 7: Smallest Kendall-Tau (KT) distance of each method over 5 runs on the polynomial ranking instances of degree 2 and 3

Method	Degree 2	Degree 3
Closest-Poly-Ranking	291	252
MaxFS	268	221
Local Search	275	277
Sampling	280	283
RankRLS	283	284
LMRC	285	295
RankSVM	291	295
Smoothed MRC	297	297
RankNet	305	304

a tractable ranking-error penalty with a weight penalty. Direct rank-metric optimization [57] and maximum-rank-correlation estimation [75] offer other approaches. Numerous polynomial extensions of these methods have been studied, such as Raykar et al. [71] extending RankSVM to quadratic functions, and SVM-rank supports polynomial kernels [43]. While many papers study the learning-to-rank problem, they do not consider optimization functions based on inversion count/Kendall's Tau. In contrast, we prove worst-case approximation guarantees for one finite input, without requiring generalization to unseen items.

Explainability of scoring/ranking functions: Feature-attribution methods quantify how an existing model's inputs contribute to its predictions. Ranking-specific methods include LIRME's local explanations [84], Rank-LIME [17], RankSHAP [18], and Ranking-SHAP [39]. We instead receive only feature vectors and a target order, without access to the model that created it. Yan et al. [89] study exact explanations that augment a linear score with hidden group bonuses. We keep the feature vectors unchanged and allow a positive optimum distance; our score has no bonus variables. Our algorithms/reductions use this fixed family of scores throughout.

Explanation of database queries [58, 72], including top-k based queries [38, 52], have been studied extensively. Roy et al. [72] design a general framework for explaining query answers through precomputed interventions. Why-not queries [38, 52] look at explanations of why an item is not part of the top-k. In addition to Shapley-based why-not queries, ShaRP [67] produces explanations for why a tuple is ranked better than another tuple. Why-not queries produce a reason about a tuple's presence in the top-k and are not adaptable to produce explanations for a complete ranking.

Gale and Marian [35] study feature-contribution metrics for *monotonic ranking functions*, whose scores do not decrease when an attribute improves. RANKHOW [15] is especially close to our setting: fit a supplied ranking by minimizing positional error. Their symbolic-gradient method searches for weight changes that improve the induced order [15]. [15] also considers inversion-based error as an optimization goal (including Kendall's Tau distance) and proposes an ILP for Kendall's Tau distance measure. We instead give approximation guarantees when the feature count grows and require the linear score to induce the full ranking without ties.

Natural language explanations for queries have been studied in different settings, such as database queries [22], provenance queries [21], etc. Producing such explanations from our results is an orthogonal problem, and ideas from Martens et al. [54] can be used to produce NL explanations.

We discuss the related work about distance functions in Appendix J.

8 CONCLUSION AND FUTURE WORK

We give approximation algorithms for explaining target rankings with linear scores. Our maximum-agreement scheme yields a randomized PTAS in logarithmic effective dimension, and the approximation factor for our minimum disagreement algorithm is governed by the effective dimension r. We also establish hardness bounds for approximation and exact optimization.

Our results raise several open questions. Can we obtain an $o(r)$-approximation for minimum disagreement, or establish a matching linear-in-r hardness bound? This question is particularly interesting when $r = O(\log n)$. Can we obtain the same maximum-agreement guarantee without random sampling, or improve its dependence on the accuracy parameters? Studying partial or top-k target orders and data that change over time is also an intriguing open direction.

It would also be interesting to explore other (non-linear) scoring families, including neural-network scores used in RankNet [9], Cobb–Douglas utilities that multiply weighted powers of attributes, and constant-elasticity-of-substitution utilities studied in database queries [47, 68]. Can we identify restrictions on these families that improve ranking fidelity while still permitting useful approximation guarantees? Finally, a natural next step is to study ranking explanations under fairness constraints. One could require representation of specified groups, regulate the attention their items receive, or protect similarly situated individuals, while minimizing deviation from the target ranking. A natural question is how these constraints change the attainable approximation guarantees.

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A MISSING DETAILS FROM SECTION 2

We use the notation of Section 2: $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ is the set of distinct items, π is the target ranking, C is its set of $M = \binom{n}{2}$ target comparisons, and Δ_e is the comparison vector for e. Recall that $r = \dim \operatorname{span}\{\Delta_e : e \in C\}$ is the effective dimension of X.

A.1 Explicit tie-breaking with rational weights

The algorithms must output a strict ranking even when a feasibility test leaves some scores equal. We prove Lemma 2.2 by giving an explicit polynomial-time construction that keeps every required comparison positive.

Proof. Assume $n \geq 2$, so every Δ_e is nonzero and $d \geq 1$. For an integer t, let $v(t) = (1, t, t^2, \ldots, t^{d-1})$. For each pair e, the function $\langle v(t), \Delta_e \rangle$ is a nonzero polynomial in t of degree at most $d-1$. It has at most $d-1$ distinct real roots. Thus all M pairs together forbid at most $M(d-1)$ integer values. Test the $M(d-1)+1$ integers from 0 through $M(d-1)$; at least one gives a vector v with no tied pair. When $d = 1$, the single choice $v = (1)$ already works. These integers and their powers have polynomial encoding length, and we test every inner product by rational arithmetic.

Let w be the supplied rational solution of $\mathcal{L}(S)$, and set $B = \max_{e \in C} |\langle v, \Delta_e \rangle|$. Define $U = 1/[2(1+B)]$. For any $0 < \alpha \leq U$ and $e \in S$, we have $\langle w + \alpha v, \Delta_e \rangle \geq 1 - UB > 1/2$. Thus the perturbation keeps all required comparisons.

For any pair, the equation $\langle w + \alpha v, \Delta_e \rangle = 0$ has at most one solution in α, because its coefficient $\langle v, \Delta_e \rangle$ is nonzero. Test the $M+1$ rational choices $\alpha_j = jU/(M+1)$, $j \in [M+1]$. Only M of them can be forbidden, so one makes all scores distinct. The arithmetic operations and rational encoding lengths remain polynomial in the input and in the supplied w. This proves both the preservation and running-time claims. For $n \leq 1$, any weight vector suffices. □

The hardness proofs also need this property for weights that may be real rather than rational. Given finitely many nonzero rational coefficient vectors, we can perturb a real weight vector so that every positive inner product stays positive and no inner product equals zero. Choose a direction whose inner product with every coefficient vector is nonzero, as in the proof of Lemma 2.2. A sufficiently small positive step in this direction preserves the finitely many positive inner products. Each coefficient vector rules out at most one step size, so a suitable size exists. The perturbation may satisfy additional inequalities, but it cannot turn a satisfied inequality into a violation.

B MISSING DETAILS FROM SECTION 3

B.1 Comparison deletion and Kendall's Tau disagreement

We prove Lemma 3.2 by relating deleted comparisons to reversed pairs. In one direction, deleting every reversed pair leaves a feasible comparison system. In the other, any feasible deletion set gives a ranking that reverses no more pairs than were deleted.

Proof. For any tie-free weight vector w, let $D_w = \{e \in C : \langle w, \Delta_e \rangle < 0\}$ denote the target comparisons reversed by σ_w. The remaining comparisons all have positive score differences. By Lemma 2.1, their comparison system is feasible, so D_w is a feasible deletion set. It has exactly $d_{\mathrm{KT}}(\pi, \sigma_w)$ elements. Applying this observation to an optimal ranking gives $\mathrm{OPT}_{\mathrm{del}} \leq \mathrm{OPT}_{\mathrm{KT}}$.

Conversely, take any feasible deletion set D and solve $\mathcal{L}(C \setminus D)$. Its solution puts every undeleted pair in the target order with score difference at least 1. It may tie some deleted pairs, but Lemma 2.2 breaks those ties while keeping every undeleted comparison positive. The resulting ranking can reverse only pairs in D, so its Kendall's Tau distance is at most $|D|$. Apply this construction to a minimum deletion set to obtain the reverse inequality. Together, the two directions prove equality. □

B.2 Implication for Spearman footrule

Spearman footrule distance charges item movements rather than inverted pairs, so the Kendall's Tau deletion analysis does not apply to it directly. The Diaconis–Graham inequalities in (2) relate the two costs. We run the same algorithm, with no change to its input or pseudocode.

Corollary B.1. *Given $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{R}^d$ and a target ranking π of X, let $r := \dim \operatorname{span}\{x_i - x_j : 1 \leq i < j \leq n\}$ be the effective dimension of X. Algorithm 2 outputs a linearly realizable ranking $\widehat{\sigma}$ in polynomial time such that $d_{\mathrm{FR}}(\pi, \widehat{\sigma}) \leq 2(r+1)\,\mathrm{OPT}_{\mathrm{FR}}(X, \pi)$. Since $r \leq n-1$, this is also a $2n$-approximation.*

Proof. For every realizable σ, the first inequality in (2) gives $d_{\mathrm{KT}}(\pi, \sigma) \leq d_{\mathrm{FR}}(\pi, \sigma)$. Taking minima over the same feasible family yields $\mathrm{OPT}_{\mathrm{KT}} \leq \mathrm{OPT}_{\mathrm{FR}}$. For the output of our algorithm, the second inequality and Theorem 3.1 give

$$d_{\mathrm{FR}}(\pi, \widehat{\sigma}) \leq 2 d_{\mathrm{KT}}(\pi, \widehat{\sigma}) \leq 2(r+1)\,\mathrm{OPT}_{\mathrm{KT}} \leq 2(r+1)\,\mathrm{OPT}_{\mathrm{FR}}\,.$$

The running time does not change, and $r + 1 \leq n$ gives the stated dimension-independent bound. □

C MISSING DETAILS FROM SECTION 4

C.1 Why one sample estimates every weight vector

Lemma 4.2 must cover a weight vector chosen *after* sampling. Estimating agreement for one fixed weight vector would not suffice: optimizing the sample can select weights that perform better on that sample than on the full comparison set. We therefore need an estimate that holds for all weight vectors at once.

Proof of Lemma 4.2. Represent the comparison vectors in $\mathbb{R}^r$, as in Section 2. Let $F(u)$ and $F_S(u)$ denote the agreement counts in these coordinates. Each weight vector accepts the comparison vectors on the positive side of a hyperplane through the origin, a set called a *homogeneous halfspace*. Formally, weight vector u accepts $H_u = \{y \in \mathbb{R}^r : \langle u, y \rangle > 0\}$. The set's *indicator* equals 1 on accepted vectors and 0 otherwise. Sample comparisons independently and uniformly from C, with replacement. The probability that H_u accepts a sampled comparison is the fraction $F(u)/M$ over all comparisons; the fraction of accepted sample occurrences is $F_S(u)/s$. Repeated comparisons count separately in this estimate.

A family *shatters* a finite set if it can accept exactly any requested subset of that set. Its *Vapnik–Chervonenkis (VC) dimension* is the largest size of a set it shatters; this measures how many acceptance choices the family can express. Homogeneous halfspaces in $\mathbb{R}^r$

have VC dimension at most r. For the upper bound, take any $r+1$ vectors y_i. They have a nonzero linear dependence $\sum_i c_i y_i = 0$. Multiply the coefficients by -1, if needed, so some $c_i > 0$. Request acceptance of all vectors with $c_i > 0$ and rejection of those with $c_i < 0$. Any realizing u would make $\sum_i c_i \langle u, y_i \rangle > 0$: the positive-coefficient terms are positive, and the negative-coefficient terms are nonnegative. This contradicts the dependence. Coefficients equal to zero do not affect the argument. Hence the set cannot be shattered.

The standard uniform estimation theorem for a family with VC dimension at most r states that, for a universal constant C,

$$s \geq C\eta^{-2}\big(r\log(2/\eta) + \log(2/\delta)\big)$$

independent samples suffice to ensure, with probability at least $1-\delta$, that every member's sample fraction differs from its true probability by at most η [4]. The theorem applies to any probability distribution, including the uniform distribution on our finite comparison list. Substituting the halfspace family gives the event $\sup_u |F_S(u)/s - F(u)/M| \leq \eta$, which is precisely (4). Weight vectors on hyperplanes cause no ambiguity: a zero inner product has indicator 0, as in the definition of F. Compression preserves all comparison signs, so the bound holds simultaneously for all original weight vectors $w \in \mathbb{R}^d$, including a vector selected after sampling. □

The conclusion holds simultaneously for a vector maximizing agreement on the sample and one maximizing agreement on all comparisons. It therefore bounds both the possible overestimate for the former vector and the possible underestimate for the latter. The main proof loses at most $2\eta M$ agreements between these two steps. Its lower bound $\mathrm{OPT}_{\mathrm{agr}} \geq M/2$ turns this additive loss into the desired multiplicative loss.

C.2 Exact optimization of the sampled comparisons

The sampling guarantee only helps if the algorithm finds the best weight vector for the sampled list. We prove Lemma 4.3 by showing that enumerating feasible subsets finds the sample optimum. We account for ties, repeated comparisons, and the number of bits needed to represent the returned weights.

Proof. The possible agreement counts form a nonempty subset of the integers $\{0, \ldots, s\}$, so some weight vector attains the largest count. If it has ties, Lemma 2.1 makes the comparison system of its positive sample pairs feasible. Rational linear programming finds a rational solution, and Lemma 2.2 preserves its required comparisons while breaking all ties. The resulting weight vector therefore agrees on every originally positive sample comparison. Its count cannot decrease, so a tie-free weight vector $w_S^\star$ that maximizes the sample objective exists.

Let $T^\star = \{t \in [s] : \langle w_S^\star, \Delta_{e_t} \rangle > 0\}$. Positive scaling makes its comparison system feasible by Lemma 2.1. When enumeration reaches $T^\star$, it finds a rational solution and breaks ties using Lemma 2.2. The candidate still agrees at every position in $T^\star$, so its count is at least $F_S(w_S^\star)$. No weight vector can exceed the sample optimum. Retaining the candidate with largest count therefore finds that optimum.

There are 2^s subsets. Each iteration solves a rational linear program, performs polynomial-time tie-breaking, and evaluates s inner products. The bounds on the number of bits used in these steps give time $2^s \operatorname{poly}(n, d, L_X, s)$. Repeated comparisons impose the same inequality, so one constraint per distinct pair suffices for feasibility. Evaluation still counts every sampled occurrence; thus removing duplicate constraints does not change the objective being optimized. □

D MISSING DETAILS FROM SECTION 5: HARDNESS PROOFS

We first prove the paired-point reduction from Section 5.1, then use it to establish the approximation lower bounds in Section 5. We also give a separate reduction for exact optimization in logarithmic ambient dimension. As in the main paper, zero and negative left-hand sides both violate a strict inequality. We count every repeated inequality separately.

D.1 The paired-point reduction

We prove the two directions of Lemma 5.1. First, we turn weights for the inequalities into a ranking that keeps each point pair consecutive. We then show how any realizable ranking gives weights for the inequalities, even if items from other pairs appear between the two items of a pair.

Proof. Let the input inequalities be $\langle a_\ell, z \rangle > 0$, $\ell \in [m]$, with nonzero rational $a_\ell \in \mathbb{Q}^q$. Create $p_\ell^+ = (m-\ell, a_\ell)$ and $p_\ell^- = (m-\ell, 0)$, in the target order $(p_1^+, p_1^-, \ldots, p_m^+, p_m^-)$. Within a pair, the points differ because $a_\ell \neq 0$. Different pairs have different first coordinates. Thus all $2m$ points are distinct, including when coefficient vectors repeat. The coordinates have polynomial encoding length.

Constructing a ranking from inequality weights. The possible violation-counts are integers between 0 and m, so a vector z attains the smallest count. Use the perturbation described in Appendix A.1 to preserve every positive inner product and make all inner products nonzero. The perturbation cannot increase the count; since z was optimal, the resulting vector is still optimal. A rational vector with the same signs also exists: require each signed inner product to be at least 1, scale the real sign realization, and solve the resulting rational linear program. Standard rational linear-programming bounds give a solution of polynomial encoding length. This existence argument does not require knowing the optimal sign pattern algorithmically.

Let $A = \max_\ell |\langle a_\ell, z \rangle|$ and set $\lambda = 2A + 1$. Use the weight vector $w = (\lambda, z)$. For pairs $\ell < h$, any point in pair ℓ has score exceeding any point in pair h by at least $\lambda(h-\ell) - 2A \geq 1$. Thus the ranking keeps each block consecutive and places the blocks in their target order. Within block ℓ, the positive point precedes the negative point exactly when $\langle a_\ell, z \rangle > 0$. Every violation therefore makes one adjacent swap. Different blocks introduce no other reversed pairs. If the violation-count is v, this ranking has Kendall's Tau distance v and footrule distance $2v$: each swapped point moves one position. Choosing weights with the minimum violation-count gives the required upper bounds.

Obtaining inequality weights from any realizable ranking. Suppose a ranking $\sigma = (b_1, \ldots, b_{2m})$ is realizable. Find a rational realizing weight vector by solving the adjacent-score constraints $\langle w, b_j - b_{j+1} \rangle \geq 1$, $j \in [2m-1]$. Their feasibility follows by scaling a strict realization; transitivity then gives the full order. The

linear program and its rational solution have polynomial size. Let $w = (\lambda, z)$ denote this solution. The inner product $\langle a_\ell, z\rangle$ is nonzero because the two points receive distinct scores. If it is negative, σ reverses that pair. These are distinct unordered pairs, so their number is at most the Kendall's Tau distance.

For the footrule bound, let a reversed pair occupy target positions $t, t+1$, and output positions u, v, respectively, with $u > v$. Its two items contribute $|u - t| + |v - (t+1)| \geq |(u - v) + 1| \geq 2$. The pairs have disjoint items, so we can add these lower bounds without counting any item's displacement twice. Hence twice the violation-count is at most the full footrule distance, even when items from different blocks appear between one another.

Every resulting violation-count is at least $\mathrm{OPT}_{\mathrm{viol}}$. Combining these lower bounds with the constructed upper bounds proves both distance identities. The agreement identity follows because agreement and Kendall's Tau disagreement add to $\binom{2m}{2}$. Solving the realization constraints and taking the last q coordinates of the resulting weights also proves the polynomial-time claim. □

D.2 Hardness from Learning Halfspaces

The approximation lower bounds follow from Learning Halfspaces. *Learning Halfspaces* asks for a linear score and a threshold that classify points with given positive or negative labels while making as few mistakes as possible. A correct prediction requires a score strictly above the threshold for a positive label and strictly below it for a negative label. Including the threshold among the unknowns turns these requirements into homogeneous strict linear inequalities.

Theorem 4.2 and Corollary 4.5 of [60] give lower bounds for this problem that apply to the homogeneous strict inequality systems defined in Section 5.1. These results rule out approximation within $S^{c_0/\log\log S}$ for some constant $c_0 > 0$ under $\mathsf{P} \neq \mathsf{NP}$, where S is the number of bits needed to describe the inequality system. Assuming the *Projection Games Conjecture* (PGC) [59], we obtain the stronger bound S^{c_1} for some constant $c_1 > 0$. Note that PGC concerns reductions from Boolean satisfiability to label-assignment problems on bipartite graphs, with small satisfiable fractions and polynomial-size label sets; Appendix D.2 states its quantitative form.

The reduction in [60] already gives homogeneous strict inequalities, so it has exactly the form required by Lemma 5.1. We state its guarantees in our notation before applying the paired-point construction. Theorem D.1 rules out approximating the minimum violation-count within factors that grow with the number of bits needed to describe the inequality system.

Theorem D.1. *Given nonzero rational vectors $a_1, \ldots, a_m \in \mathbb{Q}^q$, consider finding $z \in \mathbb{R}^q$ that minimizes the number of violated inequalities $\langle a_\ell, z\rangle > 0$. Let S be the binary description length of this system. There are constants $c_0, c_1 > 0$ such that this problem has no deterministic polynomial-time $S^{c_0/\log\log S}$-approximation unless* $\mathsf{P} = \mathsf{NP}$. *Under PGC, a deterministic polynomial-time S^{c_1}-approximation would also imply* $\mathsf{P} = \mathsf{NP}$.

Connection to classification and input size. Theorem 4.2 and Corollary 4.5 of [60] state the gaps for Learning Halfspaces. There, the error is the fraction of misclassified examples, the noise is the smallest possible error, and the failure ratio divides error by noise. Multiplying both fractions by the number of examples gives exactly the ratio of a violation-count to its optimum.

To express the classification requirements as inequalities, consider an example $u_\ell \in \mathbb{Q}^{q-1}$ with label $y_\ell \in \{-1, 1\}$. A halfspace with weight vector a and threshold b classifies this example correctly exactly when

$$\big\langle (y_\ell u_\ell, -y_\ell), (a, b)\big\rangle > 0.$$

Thus every labeled example is one homogeneous strict inequality, and a misclassification, including an example on the separating hyperplane, is exactly a violation. The coefficient vector is nonzero because its last coordinate is $-y_\ell$. Section 4.3 of the cited paper uses this homogeneous formulation and includes constraints that force its additional scaling variable to be positive.

The cited gaps are stated in terms of the size of the constructed Learning Halfspaces instance. Its list of rational coefficient vectors has a binary description length S bounded by a fixed polynomial in that size. Decreasing the positive exponent constants, if necessary, converts the published factors into the factors stated in Theorem D.1. This follows from the eventual monotonicity of $\log t/\log\log t$; the same elementary calculation is given below when we express the lower bound in the number of ranking items.

The conjecture used for the stronger gap. In a *projection game*, the input is a bipartite graph with a finite set of possible labels at each side. Every edge specifies a map from its left label set to its right label set. An assignment satisfies an edge when the right endpoint's label equals the image of the left endpoint's label. Its *value* is the fraction of satisfied edges. PGC asserts that, for some $c > 0$ and every $\gamma \geq N^{-c}$, Boolean satisfiability on input size s reduces efficiently to a game of size $N = s^{1+o(1)}\,\mathrm{poly}(1/\gamma)$ and label-set size $\mathrm{poly}(1/\gamma)$, distinguishing value 1 from value at most γ [59, Conjecture 2.2]. The weaker polynomial-size version suffices for the gap we use. We assume PGC in addition to $\mathsf{P} \neq \mathsf{NP}$.

Proof of the item-dependent bound. We prove Theorem 5.2. Given an inequality system, apply the paired-point construction. If an α-approximation for Kendall's Tau distance returns cost at most $\alpha\,\mathrm{OPT}_{\mathrm{KT}}$, Lemma 5.1 gives weights with at most $\alpha\,\mathrm{OPT}_{\mathrm{viol}}$ violations. For footrule, the returned distance is at most $2\alpha\,\mathrm{OPT}_{\mathrm{viol}}$, and the same lemma gives weights whose violation-count is at most half the returned distance. Thus either ranking approximation gives an inequality approximation with the same factor.

The paired-point instance has $n = 2m = O(S)$ items because the binary description of the inequality system contains all m inequalities. In particular, $n \leq S^2$ for all sufficiently large instances. The function $\log t/\log\log t$ is eventually increasing. Consequently,

$$\log\big(n^{c/\log\log n}\big) \leq \frac{2c\log S}{\log\log S}.$$

Choose $c < c_0/2$. The resulting approximation factor is smaller than the lower bound in Theorem D.1. Similarly, choosing $0 < c' < c_1/2$ contradicts the PGC gap because $n^{c'} \leq S^{2c'} < S^{c_1}$. These factor expressions are asymptotic; changing their values for the finitely many smaller item counts does not affect the hardness statement.

Proof of the dimension-dependent bound. We prove Theorem 5.3. Start with the paired-point data in $\mathbb{Q}^{q+1}$. Choose a rational basis matrix P of the span of all differences, and compute coordinates

y_i with $x_i - x_1 = Py_i$. The matrix has full column rank r. Every original weight w gives compressed weight $u = P^\top w$, with identical pairwise score differences. Conversely, for every $u \in \mathbb{R}^r$, the system $P^\top w = u$ has a solution because $P^\top$ has rank r. Thus compression preserves the whole family of rankings induced without ties and all three optimum values. It preserves distinctness as well: equal y_i, y_j would imply equal x_i, x_j. Gaussian elimination constructs the rational coordinates with polynomial encoding length.

The compressed differences span $\mathbb{R}^r$, so its ambient and effective dimensions coincide. Rename this ambient dimension as d. It satisfies $d = r \le q + 1 \le S + 1$. Given any returned ranking, solve its realization constraints on the original paired points and obtain inequality weights as in Lemma 5.1. We do not need the approximation algorithm to return its weights.

For sufficiently large S, use $d \le S + 1$ and eventual monotonicity to bound $d^{a/\log\log d}$ by $S^{c_0/\log\log S}$, choosing $a > 0$ sufficiently small. For PGC, choose $0 < b < c_1$; then $(S+1)^b \le S^{c_1}$ for sufficiently large S. Therefore either such dimension-dependent algorithm would give an approximation excluded by Theorem D.1. This proves the theorem with $d = r$, without relying on redundant zero coordinates. As above, the factor expressions are interpreted asymptotically; the finitely many dimensions outside the monotonic range do not affect the hardness statement.

This argument does not force $d = O(\log n)$, nor does it prove hardness for every factor $o(d)$. For example, $d/\log d$ is larger than both the stated $d^{a/\log\log d}$ factor and a sufficiently small-power d^b factor for large d. The reduction therefore leaves the gap to the main algorithm's $O(r)$ guarantee open.

D.3 Exact hardness with logarithmic ambient dimension

We next show that logarithmic dimension does not admit deterministic polynomial-time exact optimization or a deterministic FPTAS under the clique assumption stated below. In k-Clique, the input is a graph G and an integer k, and we ask whether k distinct vertices are pairwise adjacent. Under $\mathsf{FPT} \neq \mathsf{W}[1]$, no deterministic algorithm solves this problem in time $f(k)\,\mathrm{poly}(|G|)$, where $|G|$ is the number of bits needed to describe the graph [27].

We turn the choice of k pairwise adjacent vertices into linear inequalities, then into a ranking instance whose optimum determines whether the clique exists. The construction takes $f(k)\,\mathrm{poly}(|G|)$ time and ensures $d = \Theta(\log n)$. A deterministic polynomial-time exact ranking algorithm would therefore give the clique algorithm ruled out above. A deterministic FPTAS would also return the exact optimum by setting $\varepsilon = 1/(2n^2)$, because every objective value is an integer smaller than n^2. Excluding algorithms that may err with small probability requires assuming that no such clique algorithm runs in $f(k)\,\mathrm{poly}(|G|)$ time.

Theorem D.2. *Consider instances consisting of $X = \{x_1, \dots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π of X. Unless $\mathsf{FPT} = \mathsf{W}[1]$, no deterministic polynomial-time exact algorithm or deterministic FPTAS exists for minimizing Kendall's Tau distance, minimizing Spearman footrule distance, or maximizing Kendall agreement over rankings induced by linear scores, even when $d = O(\log n)$. We may require ambient dimension $d = \Theta(\log n)$.*

Preparing the graph. We may assume $k \ge 2$; smaller cases are immediate. Add isolated vertices until the graph has $N \ge \max\{2^k, k\}$ vertices, numbered $1, \dots, N$. An isolated vertex cannot enter a clique of size at least two, so the answer does not change. The resulting graph has at most $2^k + k + |G|$ vertices, and adding these vertices takes $f(k)\,\mathrm{poly}(|G|)$ time. In particular, $k \le \log N$.

Choosing vertices and checking edges. Use variables $u_1, \dots, u_k$ to choose vertex numbers, and put $\tau = 1/[4(N+2)]$. An *affine expression* is a linear expression plus a constant. For each position i and vertex a, let $h_{i,a} = u_i - a$. For each position pair $i < j$ and each ordered edge (a, b), introduce $h_{ij,ab} = u_i - a + N(u_j - b)$. Use both orientations of every undirected edge. For each expression h, include the two strict inequalities $\tau + h > 0$ and $\tau - h > 0$. There are $Q = kN + \binom{k}{2}\,2|E|$ such pairs.

Each pair has at least one satisfied inequality, and both are satisfied exactly when $|h| < \tau$. In a vertex-choice group i, at most one pair can have both inequalities satisfied because different integers a are separated by at least one and $2\tau < 1$. Likewise, in an edge group i, j, at most one pair can have both inequalities satisfied: the integers $a + Nb$, for $a, b \in [N]$, are distinct, and hence at least one apart. To check distinctness, equality would give $a - a' = N(b' - b)$; its left side has absolute value at most $N - 1$, so both differences must be zero. There are $b_k = k + \binom{k}{2}$ groups in total.

Lemma D.3 identifies when the maximum possible number of inequalities is satisfied, thereby turning the real-valued choices into a clique decision.

Lemma D.3. *No assignment satisfies more than $Q + b_k$ inequalities in the affine system. An assignment attains this value if and only if the graph contains a k-clique.*

Proof. The bound follows because each pair contributes one guaranteed satisfied inequality and each group contributes at most one additional satisfied inequality. If vertices $a_1, \dots, a_k$ form a clique, choose $u_i = a_i$. Each vertex-choice group has a zero expression, and each edge group contains the ordered edge (a_i, a_j) with zero expression. Consequently, each of the b_k groups contributes one additional satisfied inequality.

Conversely, attaining the bound requires some pair in every group to have both inequalities satisfied. Let a_i be the vertex associated with that pair in group i, so $|u_i - a_i| < \tau$. For each $i < j$, let (a, b) be the ordered edge whose group expression has absolute value less than τ. The integer $a_i - a + N(a_j - b)$ has absolute value less than $(N+2)\tau = 1/4$, using the two vertex-choice bounds and the edge bound. The integer must therefore be zero. The equality $a_i - a = N(b - a_j)$, together with $|a_i - a| \le N - 1$, forces $a_i = a$ and $a_j = b$. Thus the chosen vertices have an edge between every pair. The graph has no self-loops, so they are distinct and form a clique. □

Making every inequality homogeneous. Introduce one additional variable t. Replace every constant by that constant times t: for vertex groups, use $h = u_i - at$, and for edge groups use $h = u_i + Nu_j - (a + Nb)t$. Replace each pair by $\tau t + h > 0$ and $\tau t - h > 0$, and add the inequality $t > 0$. The resulting system has $q = k + 1$ variables and $m = 2Q + 1$ inequalities, all with nonzero rational coefficient vectors. Lemma D.4 shows that the additional inequality prevents the new variable from changing the intended threshold.

Lemma D.4. *The homogeneous system has at most $B = Q - b_k$ violations if and only if the graph contains a k-clique. If the graph contains a k-clique, the optimum is exactly B; otherwise, it is at least $B + 1$.*

Proof. If $t > 0$, divide all variables by t. The first $2Q$ inequalities become the affine system in Lemma D.3, and the additional inequality is satisfied. Their maximum possible satisfaction count is $Q + b_k + 1$, giving at least $(2Q + 1) - (Q + b_k + 1) = B$ violations. Equality occurs exactly when there is a clique.

If $t \le 0$, the two left sides in each pair sum to $2\tau t \le 0$, so they cannot both be positive. At most Q inequalities are satisfied, and $t > 0$ also fails. There are then at least $Q + 1 > B$ violations. This excludes all unintended values of t. Finally, $Q \ge kN \ge b_k$, so B is nonnegative. The objectives are integers, yielding the claimed one-unit gap. □

Transferring the threshold to rankings. Apply the paired-point reduction. It creates $n = 4Q+2$ distinct points in ambient dimension $d = k + 2$. The clique answer is yes exactly when $\mathrm{OPT}_{\mathrm{KT}} \le B$, equivalently $\mathrm{OPT}_{\mathrm{FR}} \le 2B$, equivalently $\mathrm{OPT}_{\mathrm{agr}} \ge \binom{n}{2} - B$. The coefficients have polynomial encoding length in k and $\log N$, and $Q = O(k^2N^2)$. Together with the padding, the entire construction takes $f(k)\,\mathrm{poly}(|G|)$ time and output size. Moreover, $d = k + 2 \le \log N + 2 = O(\log n)$. A polynomial-time exact algorithm for any of the ranking objectives under this dimension restriction would therefore solve k-Clique in the excluded parameterized time.

To require $d = \Theta(\log n)$, append $\lceil \log n \rceil$ zero coordinates to every point. No score difference or realizable ranking changes. The new ambient dimension lies between $\log n$ and $O(\log n)$, and the output size remains bounded by $f(k)\,\mathrm{poly}(|G|)$. This padding controls ambient dimension only; it does not assert that the effective dimension is $\Theta(\log n)$.

Why an FPTAS would give exact optimization. Suppose an FPTAS exists for one of the restricted objectives. Run it with $\varepsilon = 1/(2n^2)$. Every feasible Kendall's Tau distance and agreement value is at most $\binom{n}{2} < n^2$, and every footrule distance is at most $n(n-1) < n^2$. For minimization, the returned cost exceeds optimum by less than $\varepsilon n^2 = 1/2$. For maximization, the optimum minus the returned value is less than $1/2$. The algorithm returns a realizable ranking with integer objective, so either guarantee forces exact optimality. Its running time is polynomial in the input size and $1/\varepsilon = 2n^2$, hence polynomial overall, contradicting the exact lower bound. For randomized schemes this argument rules out schemes with an always-correct guarantee; excluding bounded-error randomized exact algorithms (correct with probability at least $2/3$) requires the corresponding randomized clique assumption. Our exact and FPTAS hardness statements use the standard deterministic convention.

E ADDITIONAL HARDNESS UNDER ALTERNATIVE ASSUMPTIONS

The paired-point construction also transfers classical lower bounds for minimizing the number of unsatisfied linear inequalities. These bounds are weaker than those in Section 5, but give a constant-factor lower bound under $\mathsf{P} \neq \mathsf{NP}$ and a logarithmic lower bound under a different complexity assumption. Both concern arbitrary dimension, not logarithmic dimension.

The *minimum unsatisfied homogeneous strict linear relations* problem asks for a vector that violates as few input inequalities as possible. More specifically, an instance of $\textsc{Min-ULR}^{>}_{\mathrm{hom}}$ consists of nonzero $a_1, \ldots, a_m \in \{-1, 0, 1\}^q$ and minimizes the number of violations of $\langle a_\ell, z\rangle > 0$. Amaldi and Kann prove the following approximation lower bounds [3]. The condition $m = 2q$ lets us express the logarithmic factor in the number of ranking items without losing its leading constant.

Theorem E.1. *Given nonzero vectors $a_1, \ldots, a_m \in \{-1, 0, 1\}^q$, let $\textsc{Min-ULR}^{>}_{\mathrm{hom}}$ minimize the number of violated inequalities $\langle a_\ell, z\rangle > 0$ over $z \in \mathbb{R}^q$. This problem has no deterministic polynomial-time constant-factor approximation unless $\mathsf{P} = \mathsf{NP}$. For every fixed $0 < \varepsilon < 1$, it has no deterministic polynomial-time $(1 - \varepsilon)\ln q$-approximation unless $\mathsf{NP} \subseteq \mathsf{DTIME}(q^{O(\log\log q)})$. The hard instances use $m = 2q$.*

Deriving the inequality system from a graph. The reduction starts from a graph with q vertices. A *dominating set* is a vertex subset such that every vertex is in the subset or has a neighbor in it. The *closed neighborhood* of a vertex consists of that vertex and its neighbors; let $N[i]$ denote this set for vertex i. For each vertex i, use the two inequalities $-z_i > 0$ and $\sum_{j \in N[i]} z_j > 0$. These are $2q$ homogeneous strict inequalities, each with a nonzero coefficient vector whose entries belong to $\{-1, 0, 1\}$.

If D is a dominating set, assign $z_i = q$ for $i \in D$ and $z_i = -1$ otherwise. Every neighborhood sum is positive, because it contains a selected vertex and at most $q - 1$ negative terms. Exactly $|D|$ inequalities of the first kind fail. Conversely, given any vector, let $D_0 = \{i : z_i \ge 0\}$ and add to it every vertex whose closed neighborhood misses D_0. Each added vertex has an entirely negative neighborhood, so its sum inequality fails. Each member of D_0 fails its first inequality. The resulting set dominates the graph and its size is at most the violation-count. This proves equality of the optima and preserves every approximation factor. The dominating-set lower bounds used by Amaldi and Kann therefore give the stated theorem. Here ln denotes the natural logarithm, and $\mathsf{DTIME}(g(q))$ denotes decision problems solvable by a deterministic algorithm in time $g(q)$. The stronger complexity assumption excludes time of the form $q^{O(\log\log q)}$ for all NP problems, not merely ordinary polynomial time.

We now apply Lemma 5.1 to this inequality system. Theorem E.2 states the resulting bounds under these alternative assumptions, separately from the stronger Learning Halfspaces lower bounds.

Theorem E.2. *Consider pairwise distinct vectors $x_1, \ldots, x_n \in \mathbb{Q}^d$, let $X = \{x_1, \ldots, x_n\}$, and let π be a target ranking of X. The dimension d is part of the input. Unless $\mathsf{P} = \mathsf{NP}$, the minimum-disagreement problems under Kendall's Tau and Spearman footrule distances admit no deterministic polynomial-time constant-factor approximation. For every fixed $0 < \varepsilon < 1$, neither admits a deterministic polynomial-time $(1 - \varepsilon)\ln n$-approximation unless $\mathsf{NP} \subseteq \mathsf{DTIME}(n^{O(\log\log n)})$. These bounds hold with $d = \Theta(n)$.*

Proof. Apply the paired-point construction to the homogeneous strict inequality system. A constant-factor Kendall's Tau approximation would give weights with the same approximation factor for the violation-count. For footrule, the optimum is twice the minimum violation-count, and Lemma 5.1 gives weights with at most half as many violations as the returned distance, preserving

the factor again. Both conversions take polynomial time, contradicting Theorem E.1. The inequality system has $m = 2q$, so the ranking instance has $n = 4q$ and $d = q + 1 = \Theta(n)$.

For the logarithmic factor, fix ε and use the inequality bound with $\varepsilon/2$. For sufficiently large q, $(1-\varepsilon)\ln(4q) \le (1-\varepsilon/2)\ln q$, because the additive term $(1-\varepsilon)\ln 4$ is eventually smaller than $(\varepsilon/2)\ln q$. The resulting approximation factor would therefore contradict Theorem E.1. Replacing q by $n = 4q$ also preserves the stated time class, which proves the second claim. □

F MISSING DETAILS FROM SECTION 7: RELATED WORK

Our problem connects ranked query processing, learning to rank, ranking explanation, rank distances, and linear inequalities. We seek one common score on the supplied features that approximates a complete target order. The following discussion explains how this objective differs from those studied in each of these areas.

F.1 Ranked retrieval and utility queries

Fagin et al. developed algorithms for combining scores through sorted and random access [32], and surveyed the broader problem of combining fuzzy information [30]. These methods efficiently evaluate an aggregation rule; they do not choose a rule to reproduce an externally supplied order. The onion technique uses geometric layers to index linear optimization queries [13]. RankingCube organizes multidimensional data to answer ranking queries [88], while Das et al. answer top-k queries by combining information from materialized views [19]. Reverse top-k queries ask which weight vectors place a designated tuple among the top results [85]. All these formulations connect weights to ranked query answers. Our objective instead evaluates the entire induced order against a target by counting reversed pairs or summing changes in item positions.

Other work addresses uncertainty about user preferences. A *skyline* contains tuples for which no other tuple is at least as good in every relevant attribute and strictly better in at least one [8]. *Regret-minimizing representative sets* select a small subset whose best item remains close to the best item in the full data across a family of utility functions [62]. *Rank-regret representatives* measure this loss by rank rather than utility [7]; Shetiya et al. give a unified optimization approach for several representative-set problems [76]. These methods choose items, whereas we keep every item and choose the scoring rule.

Interactive approaches obtain additional preference information from the user. Nanongkai et al. study interactive regret minimization [61]; Qian et al. learn preferences through adaptive pairwise comparisons [69]; Wang et al. search interactively for one of a user's top-k items [86]. Our input already specifies a full order. Our objective measures how closely a fixed linear feature representation can realize that order; it neither selects follow-up questions nor returns a subset of items.

F.2 Learning and explaining linear rankings

Learning-to-rank methods estimate scores from relevance judgments, clicks, or pairwise preferences. Joachims uses clickthrough data to learn pairwise ranking preferences with a support-vector method [42]. RankNet learns from pairwise comparisons using a differentiable probabilistic loss [9]. Sculley studies scalable learning-to-rank methods [74], and specialized algorithms improve the efficiency of linear ranking support-vector machines [2, 49]. These methods optimize training losses rather than the exact number of reversed pairs for one complete target order. For example, RankSVM penalizes both comparison errors and the size of the weight vector, while RankNet uses a smooth probabilistic loss.

Classical ordinal regression fits ordered responses through linear programming [81]. Metzler et al. directly optimize rank-based retrieval metrics using a numerical search over scoring parameters [57]. Shen et al. study linearized maximum-rank-correlation estimation [75]. These works motivate optimizing ranking quality directly, but their statistical or numerical guarantees differ from our worst-case approximation guarantees for a fixed finite input. We do not claim that optimizing a smooth training loss solves the minimum-disagreement problem.

A second line of work explains rankings or synthesizes scores. Gale and Marian study feature-contribution metrics for *monotonic ranking functions*, whose scores do not decrease when an attribute improves [35]. RANKHOW is especially close to our setting: Chen et al. fit a supplied ranking by minimizing positional error, allow weight constraints and score ties, and give an optimization model that combines real-valued weights with discrete ordering decisions. Their symbolic-gradient method searches for weight changes that improve the induced order [15]. Their exact region-enumeration bound has an exponent that depends on the number of features, so its polynomial-time conclusion assumes a fixed feature count. They also discuss inversion-based errors, including Kendall's Tau distance; their principal optimization formulation uses positional error. We instead give approximation guarantees when the feature count grows and require the linear score to induce the full ranking without ties. Why-not analyses identify changes that would place a missing item in a top-k answer or explain reverse top-k query results [38, 52]. RANKHOW directly optimizes ranking error; the why-not formulations focus on a designated item or subset. Our results complement these systems with worst-case guarantees and dimension-dependent lower bounds.

Feature-attribution methods quantify how an existing model's inputs contribute to its predictions. Ranking-specific methods include LIRME's local explanations [84], Rank-LIME [17], RankSHAP [18], and RankingSHAP [39]. We instead receive only feature vectors and a target order, without access to the model that created it. Yan et al. study exact explanations that augment a linear score with hidden group bonuses [89]. We keep the feature vectors unchanged and allow a positive optimum distance; our score has no bonus variables. Our algorithms and reductions use this fixed family of scores throughout.

F.3 Polynomial scores and nonlinear utilities

Polynomial scoring has a long history in information retrieval. Fuhr maps query–document pairs to feature vectors and fits polynomial estimates of relevance probabilities by minimizing squared error [34]. This estimates a numerical response from relevance feedback. Our polynomial extension instead receives a complete target order and optimizes its pairwise or positional disagreement; it needs neither relevance probabilities nor target score values.

Polynomial kernels provide another route to nonlinear ranking models. A *kernel* evaluates an inner product in a transformed feature representation without explicitly constructing its coordinates. More specifically, Joachims's SVM-rank documentation supports the polynomial kernel $K(x, z) = (s\langle x, z\rangle + c)^p$, with configurable degree p, scale s, and offset c [43]. Expanding this expression gives products of feature coordinates; with positive s, c, its feature representation includes the monomials up to degree p, with fixed rescaling. Raykar et al. explicitly compare linear RankSVM with a quadratic variant using $K(x, z) = (\langle x, z\rangle + c)^2$ in their experiments [71]. Their main algorithm uses a differentiable lower bound on a generalized Wilcoxon–Mann–Whitney statistic, which measures how often items from ordered groups appear in the correct relative order. The quadratic RankSVM is a benchmark, not a theorem giving an approximation ratio for fitting a ranking with an unrestricted quadratic score. Ranking support-vector methods trade a penalty on comparison errors against a penalty on the score parameters. These training objectives differ from minimizing the exact ranking distance over all polynomial coefficients on a fixed input.

The database literature also considers nonlinear utility functions. Kessler Faulkner et al. study utilities that model diminishing returns, risk preferences, and substitution between attributes [47]. Qi et al. study multiplicative utilities, including Cobb–Douglas and constant-elasticity-of-substitution families [68]. Cobb–Douglas utilities multiply weighted powers of positive attributes; constant-elasticity-of-substitution utilities use a power aggregate to model substitution between attributes. These papers choose a small representative subset whose best item loses little utility relative to the best item in the full database. We keep every item and choose a common score to approximate the entire supplied order. Their regret bounds therefore address a different output and objective from our ranking-distance guarantees.

Our use of monomial features is standard, and we do not claim a new polynomial scoring family. Chen et al. also explicitly discuss polynomial-derived attributes for RANKHOW [15]. Appendix G uses that representation to prove deterministic $(r_p + 1)$- and $2(r_p + 1)$-approximations for minimum Kendall's Tau and footrule disagreement, respectively, where r_p is the effective dimension of the expanded features. It also proves a randomized maximum-agreement scheme that is FPT in r_p and the accuracy parameters, and a PTAS when $r_p = O(\log n)$. These are multiplicative guarantees relative to the best unrestricted polynomial score of each fixed maximum degree, rather than guarantees for relevance estimation or a regularized training loss.

The hardness extension needs more than feature expansion. Every linear score is a polynomial score, but allowing higher-degree terms may improve the optimum on an instance that is hard for linear scores. Our separate block construction converts *any* ranking induced by a polynomial score into weights for the original inequalities. This conversion increases the approximation factor by at most a constant that depends on the fixed degree. Theorems G.7 and G.8 consequently retain strong minimum-disagreement inapproximability as a function of item count, original dimension, and expanded effective dimension. These are computational lower bounds for fitting a complete order, not lower bounds on the utility loss of a representative subset. They concern arbitrary dimension; we do not claim that this reduction extends Theorem D.2 to higher degrees in logarithmic dimension.

F.4 Rank distances and aggregation

Kendall's Tau distance counts reversed pairs, while Spearman footrule adds the positional movements of the items. Diaconis and Graham show that footrule distance lies between Kendall's Tau distance and twice that distance [24]. We use this theorem only to transfer the approximation upper bound. Our paired-point construction proves an exact factor-two identity for its own instances; that identity is stronger than what the general metric comparison alone would give.

Rank aggregation combines several input orders into a consensus. Dwork et al. study aggregation for Web rankings [28], and Davenport and Kalagnanam investigate algorithms for the Kemeny rule, which minimizes pairwise disagreement with a collection of orders [20]. We receive one target, not a collection of votes. Moreover, we cannot choose an arbitrary permutation: the feature vectors restrict which orders a common linear score can produce. The difficulty therefore comes from realizability, not disagreement among several targets. Complementing the pair count equates exact maximum agreement and minimum Kendall's Tau disagreement, but does not equate their multiplicative approximation guarantees.

F.5 Geometry, feasible comparisons, and hardness

The weights that tie two items form a flat boundary in weight space, called a *hyperplane*. Formally, this boundary satisfies $\langle w, x_i - x_j\rangle = 0$. The collection of these boundaries forms a *hyperplane arrangement*. Removing them divides the remaining space into connected regions, called *cells*; every pair keeps its sign within a cell, so the ranking stays unchanged. This geometric description is classical [29, 70, 82]. Work on linear programming in fixed or low dimension provides another reason to treat dimension as a structural parameter [12, 56]. Yan et al. [89] show a transformation of the Kendall Tau optimization problem into a search over the cells of the arrangement of $\binom{n}{2}$ hyperplanes in d-dimensional weight space. Our algorithms do not enumerate all these regions. The minimum-disagreement algorithm uses small incompatible sets of comparisons. The agreement scheme samples comparisons and uses a uniform estimation theorem for linear comparisons [4]; Appendix C.1 explains the relevant terms.

Orienting comparisons by the target gives strict homogeneous linear inequalities. Deleting as few inequalities as possible to obtain feasibility is equivalent to minimizing Kendall's Tau disagreement. This connects our objective to maximum feasible subsystems and minimum unsatisfied linear relations [3, 16]. The ranking setting adds structure: comparisons arise from all pairs of one point set, and the output must be a full strict ranking. Our reduction explicitly preserves this structure and proves exact relations for both distances and agreement. The strongest approximation lower bounds use the homogeneous inequality construction of [60], with the stronger exponents conditional on the Projection Games Conjecture [59]. For logarithmic ambient dimension, Appendix D.3 gives a separate reduction from k-Clique; the standard parameterized-complexity framework explains the assumption used in this result [27].

G MISSING DETAILS FROM SECTION 1.1: POLYNOMIAL SCORES

Polynomial scores can capture interactions between attributes that a linear score cannot express. For example, a term $x_1^2 x_2$ lets the contribution of one attribute depend on another. By treating monomials as additional features, we extend both the maximum-agreement scheme from Section 4 and the minimum-disagreement approximations from Section 3 to every fixed degree. The approximation guarantees depend on the effective dimension of these expanded features. We first define this representation and prove the approximation guarantees, including the quadratic case. We then give a separate reduction that establishes minimum-disagreement hardness for every fixed degree.

G.1 Polynomial scores and their feature representation

Fix a positive integer p, the maximum degree allowed; it is a constant, not part of the input. A *monomial* multiplies feature coordinates raised to nonnegative integer powers, and its *total degree* is the sum of those powers. More specifically, for an exponent tuple $\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{Z}^d_{\geq 0}$, let $x^\alpha = \prod_{j=1}^d x_j^{\alpha_j}$ and $|\alpha| = \sum_{j=1}^d \alpha_j$. A factor with exponent zero equals 1, including when its coordinate is zero. Let $\mathcal{I}_p = \{\alpha : 1 \leq |\alpha| \leq p\}$. An *unrestricted polynomial score of degree at most p* combines these monomials with arbitrary real coefficients, including negative ones. Since rankings depend only on score differences, we omit the constant monomial and let

$$f_a(x) = \sum_{\alpha \in \mathcal{I}_p} a_\alpha x^\alpha.$$

The *monomial feature map* lists every nonconstant monomial of degree at most p. Formally, let $\Phi_p(x) = (x^\alpha)_{\alpha \in \mathcal{I}_p}$, using the same order of exponent tuples for all items. Its number of coordinates is $D_p = \binom{d+p}{p} - 1$. The *expanded effective dimension*, denoted by r_p, is the dimension spanned by the differences $\Phi_p(x_i) - \Phi_p(x_j)$, over $i < j$. Thus $r_p \leq \min\{D_p, n-1\}$. We optimize over the rankings induced without ties by these polynomial scores. Let $\mathrm{OPT}^{(p)}_{\mathrm{agr}}$ denote their largest Kendall agreement with π, and let $\mathrm{OPT}^{(p)}_{\mathrm{KT}}$ and $\mathrm{OPT}^{(p)}_{\mathrm{FR}}$ denote their smallest Kendall's Tau and Spearman footrule distances from π, respectively.

Table 8 summarizes the polynomial-scoring approximation guarantees and hardness bounds.

G.2 Approximating maximum agreement

Applying the linear agreement scheme to the expanded features gives the same approximation ratio and success probability for polynomial scores. Theorem G.1 gives this guarantee and includes the cost of computing the monomials and representing them as rational numbers. This cost is polynomial for every fixed degree. Let poly_p denote a polynomial whose constants and exponent may depend on p, but not on the effective dimension or the accuracy parameters. A subscript p on O_p likewise allows the hidden constant to depend on the fixed degree.

Theorem G.1. *Fix an integer $p \geq 1$, and let $n \geq 2$. For pairwise distinct vectors $x_1, \ldots, x_n \in \mathbb{Q}^d$, let $X = \{x_1, \ldots, x_n\}$, and let π be a target ranking of X. Let r_p be the dimension of the span of the vectors $\Phi_p(x_i) - \Phi_p(x_j)$ over all $1 \leq i < j \leq n$. There exists a randomized algorithm that, given X, π, and $0 < \varepsilon, \delta < 1$, outputs a polynomial score of total degree at most p with rational coefficients and pairwise distinct item scores, and runs in time*

$$2^{O((r_p \log(2/\varepsilon) + \log(2/\delta))/\varepsilon^2)} \, \mathrm{poly}_p(n, d, L_X).$$

With probability at least $1 - \delta$, its induced ranking $\widehat{\sigma}$ satisfies $\mathrm{agr}_{\mathrm{KT}}(\pi, \widehat{\sigma}) \geq (1 - \varepsilon)\,\mathrm{OPT}^{(p)}_{\mathrm{agr}}(X, \pi)$. For fixed p, the scheme is FPT jointly in r_p, $1/\varepsilon$, and $\log(1/\delta)$. For fixed failure probability, it is a randomized PTAS when $r_p = O(\log n)$; the original dimension bound $d = O((\log n)^{1/p})$ suffices. If $d = O(\log n)$, then for fixed p, ε, δ its running time is $2^{O_p((\log n)^p)}\,\mathrm{poly}_p(n, L_X)$.

The algorithm has three steps. First, enumerate the exponent tuples and compute $\Phi_p(x_i)$ for every item. Next, run Algorithm 3 on the expanded vectors with the original target order and the requested accuracy and failure probability. Finally, use its returned weight on each expanded feature as the coefficient of the corresponding monomial. The resulting polynomial orders the original items exactly as the returned linear score orders their expanded vectors.

Proof. We first bound the size of the expanded instance. To count all monomials of degree at most p, including the constant, assign each exponent tuple an extra entry $p - |\alpha|$. The resulting $d+1$ nonnegative entries sum to p. An arrangement of p identical degree units and d separators specifies exactly one such tuple: the separators divide the units among the $d+1$ entries. Choosing their positions gives $\binom{d+p}{p}$ tuples. Removing the constant monomial gives the claimed D_p. For fixed p, this is $O_p(d^p)$. We can enumerate the tuples directly: for each degree $t \in \{1, \ldots, p\}$, list the nondecreasing sequences of t feature indices from $[d]$. Counting the occurrences of each index gives exactly one exponent tuple. Every monomial appears once, so this enumeration requires polynomial work per tuple.

Each expanded coordinate multiplies at most p input coordinates, counting repetitions. If an input rational has at most L_X bits, the numerator and denominator of such a product have $O(p(L_X + 1))$ bits before reduction; reducing the fraction cannot increase this bound. We compute these products exactly by rational arithmetic. Enumerating the exponent tuples and computing all nD_p coordinates therefore takes $\mathrm{poly}_p(n, d, L_X)$ time. The maximum encoding length of one expanded coordinate, denoted by L_{Φ_p}, is also bounded by such a polynomial.

We next prove that expansion preserves the feasible rankings. Every polynomial coefficient vector a gives the linear score $\langle a, \Phi_p(x) \rangle = f_a(x)$. Conversely, every linear score on the expanded coordinates gives a polynomial of degree at most p on the original features. In particular, for every pair of items,

$$f_a(x_i) - f_a(x_j) = \langle a, \Phi_p(x_i) - \Phi_p(x_j) \rangle.$$

The two representations give identical score differences, so they produce the same ties and the same rankings induced without ties. Since $p \geq 1$, the expanded vectors include every original coordinate, so distinct original items remain distinct. We can therefore apply the linear algorithm. This equality of the ranking families also shows that the optimal linear agreement on the expanded data equals $\mathrm{OPT}^{(p)}_{\mathrm{agr}}$.

Table 8: Polynomial-scoring results for Kendall's Tau disagreement and agreement, for any fixed maximum degree $p \geq 1$. All approximation algorithms run in polynomial time; the PTAS is randomized. Here $r_p \leq \binom{d+p}{p} - 1$ is the expanded effective dimension. Hardness exponents may depend on p, and dimension-dependent bounds also hold with r_p in place of d.

Objective	Setting	Guarantee or lower bound	Condition
Min disagreement	arbitrary r_p	$(r_p + 1)$-approximation (Theorem G.2)	–
Max agreement	$r_p = O(\log n)$	PTAS (Theorem G.1)	–
Min disagreement	arbitrary dimension	no $n^{c_p/\log\log n}$- or $d^{a_p/\log\log d}$-approximation (Theorems G.7 and G.8)	P ≠ NP
Min disagreement	arbitrary dimension	no $n^{c'_p}$- or d^{b_p}-approximation (Theorems G.7 and G.8)	PGC, P ≠ NP
Both objectives	arbitrary dimension	no deterministic exact algorithm or FPTAS (Corollary G.9)	P ≠ NP

Now run Algorithm 3 on the expanded instance. Theorem 4.4 applies with ambient dimension D_p, effective dimension r_p, and encoding length L_{Φ_p}. Its tie-breaking step gives a rational coefficient vector $\widehat{a}$ with distinct scores on all expanded items. The polynomial with these coefficients gives the same distinct scores and hence the same ranking $\widehat{\sigma}$ on the original items. With probability at least $1 - \delta$, its agreement is at least $1 - \varepsilon$ times the expanded linear optimum. Equality of the two optima proves the claimed approximation guarantee.

The linear algorithm has the exponential factor stated in the theorem and a remaining factor $\mathrm{poly}(n, D_p, L_{\Phi_p})$. Our bounds on D_p and L_{Φ_p} turn its polynomial factor, together with the cost of feature expansion, into $\mathrm{poly}_p(n, d, L_X)$. This proves the stated running time and the FPT claim for each fixed p.

If $r_p = O(\log n)$, the exponential factor is a fixed power of n for fixed ε, δ, so the scheme is a PTAS. Since $r_p \leq D_p = O_p(d^p)$, the bound $d = O((\log n)^{1/p})$ ensures this condition. If instead $d = O(\log n)$, then $r_p = O_p((\log n)^p)$. Substituting into the running time gives the last claim. For $p \geq 2$, this bound need not be polynomial: the exponent on n may grow as $(\log n)^{p-1}$. □

The polynomial running-time bound requires a fixed degree. If p grows with the input, the number D_p of monomials can grow faster than any polynomial in the original input size. The equality of the ranking families still holds, but the proof does not give polynomial-time feature expansion or an FPT guarantee jointly in the degree and the other parameters.

G.3 Approximating minimum disagreement

The same representation lets us approximate the number of reversed target pairs. Unlike the agreement scheme, the following algorithm needs neither random sampling nor a bound on the effective dimension to run in polynomial time. Increasing the degree can reduce the smallest achievable disagreement, but it can also increase the approximation factor through r_p.

Theorem G.2. *Fix an integer $p \geq 1$, and let $n \geq 2$. For pairwise distinct vectors $x_1, \ldots, x_n \in \mathbb{Q}^d$, let $X = \{x_1, \ldots, x_n\}$, and let π be a target ranking of X. Let r_p be the dimension of the span of the vectors $\Phi_p(x_i) - \Phi_p(x_j)$ over all $1 \leq i < j \leq n$, and let $D_p := \binom{d+p}{p} - 1$ be the number of expanded features. There exists a deterministic algorithm that, given X and π, outputs a polynomial score of total degree at most p with rational coefficients and pairwise distinct item scores, and runs in time $\mathrm{poly}_p(n, d, L_X)$. Its induced ranking $\widehat{\sigma}$ satisfies $d_{\mathrm{KT}}(\pi, \widehat{\sigma}) \leq (r_p + 1)\,\mathrm{OPT}_{\mathrm{KT}}^{(p)}(X, \pi)$. The factor is at most $D_p + 1 = \binom{d+p}{p}$ and at most n.*

To implement the algorithm, compute the expanded vectors $\Phi_p(x_i)$ and run Algorithm 2 on them in the original target order. Each comparison now requires the weighted sum of monomial differences to be positive; the weights themselves remain unrestricted. Once Algorithm 2 finds a feasible subset of the target comparisons, it computes rational weights and breaks ties while preserving that subset. Use each resulting weight as the coefficient of its monomial. Sorting the polynomial scores gives the output. Thus the algorithm works in the expanded feature space and does not require a separate search over polynomial functions.

Proof. The construction in the proof of Theorem G.1 computes all expanded vectors in $\mathrm{poly}_p(n, d, L_X)$ time. Their ambient dimension is D_p, their effective dimension is r_p, and their maximum coordinate encoding length L_{Φ_p} is polynomial in the original input for fixed p. They remain pairwise distinct because their degree-one coordinates include all the original features.

For every coefficient vector a, the identity $f_a(x_i) - f_a(x_j) = \langle a, \Phi_p(x_i) - \Phi_p(x_j)\rangle$ holds for every pair. Conversely, every linear weight vector on the expanded coordinates defines an allowed polynomial score. Hence the linear rankings of the expanded items and the polynomial rankings of the original items form exactly the same family of orders. The target order also stays the same, so each corresponding ranking has the same Kendall's Tau distance. In particular, the minimum linear distance on the expanded instance equals $\mathrm{OPT}_{\mathrm{KT}}^{(p)}$.

Apply Algorithm 2 to this expanded instance. Theorem 3.1 applies with effective dimension r_p and gives the claimed factor. More explicitly, Lemma 3.3 bounds each removed conflict by $r_p + 1$ comparisons. Lemma 3.5 bounds the number T of these disjoint conflicts by the optimal deletion cost, which equals $\mathrm{OPT}_{\mathrm{KT}}^{(p)}$ by Lemma 3.2 and the equality of the ranking families. If D is the union of the removed conflicts, then

$$d_{\mathrm{KT}}(\pi, \widehat{\sigma}) \leq |D| \leq (r_p + 1)T \leq (r_p + 1)\,\mathrm{OPT}_{\mathrm{KT}}^{(p)}.$$

The first inequality holds because tie-breaking preserves all comparisons outside D. Use the final rational weights as polynomial coefficients. The resulting polynomial gives the same distinct scores and therefore the same pairwise orders.

The linear algorithm uses a polynomial number of rational linear-programming tests and a polynomial-time tie-breaking step. Its running time on the expanded input is therefore $\mathrm{poly}(n, D_p, L_{\Phi_p})$. The bounds on the expansion turn this into $\mathrm{poly}_p(n, d, L_X)$, including the cost of computing the monomials and outputting the rational coefficients. There is no restriction on d or r_p here: only p is fixed. Finally, $r_p \le \min\{D_p, n-1\}$ gives both stated factor bounds. □

The same output also approximates the minimum Spearman footrule distance. The ranking that minimizes footrule need not minimize Kendall's Tau distance, but the Diaconis–Graham inequalities relate the two optima over the same polynomial scoring family. Corollary G.3 gives the resulting guarantee without an additional optimization step.

Corollary G.3. *For every fixed integer $p \ge 1$, there exists a deterministic algorithm that, given $n \ge 2$ pairwise distinct vectors $X = \{x_1, \dots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π, outputs a polynomial score whose induced ranking $\widehat{\sigma}$ satisfies $d_{\mathrm{FR}}(\pi, \widehat{\sigma}) \le 2(r_p+1)\,\mathrm{OPT}_{\mathrm{FR}}^{(p)}(X, \pi)$, where*

$$r_p := \dim \mathrm{span}\{\Phi_p(x_i) - \Phi_p(x_j) : 1 \le i < j \le n\}.$$

The algorithm runs in time $\mathrm{poly}_p(n, d, L_X)$. Its approximation factor is also at most $2\binom{d+p}{p}$ and at most $2n$.

Proof. Every ranking induced without ties by a polynomial score has Kendall's Tau distance from π at most its footrule distance, by (2). Taking minima over the same polynomial ranking family yields $\mathrm{OPT}_{\mathrm{KT}}^{(p)} \le \mathrm{OPT}_{\mathrm{FR}}^{(p)}$. For the output of Theorem G.2, the other Diaconis–Graham inequality gives

$$\begin{aligned} d_{\mathrm{FR}}(\pi, \widehat{\sigma}) &\le 2 d_{\mathrm{KT}}(\pi, \widehat{\sigma}) \\ &\le 2(r_p + 1)\,\mathrm{OPT}_{\mathrm{KT}}^{(p)} \\ &\le 2(r_p + 1)\,\mathrm{OPT}_{\mathrm{FR}}^{(p)}. \end{aligned}$$

We reuse the same coefficients, tie-breaking, and ranking, so the algorithm remains deterministic and its running time does not change. The bounds $r_p + 1 \le D_p + 1$ and $r_p + 1 \le n$ give the remaining claims. □

G.4 Quadratic scores as a special case

For $p = 2$, we have $D_2 = d(d+3)/2$: the expanded vector contains the original features, their squares, and the products of distinct feature pairs. Theorem G.2 and Corollary G.3 give deterministic $(r_2 + 1)$- and $2(r_2 + 1)$-approximations for the two disagreement distances. Their factors are at most $d(d+3)/2 + 1$ and $d(d+3) + 2$, respectively, and also at most n and $2n$. Corollary G.4 gives the maximum-agreement guarantee for quadratic scores and identifies when its running time is polynomial.

Corollary G.4. *Let $X = \{x_1, \dots, x_n\} \subseteq \mathbb{Q}^d$ be a set of pairwise distinct vectors, let π be a target ranking of X, and let*

$$r_2 := \dim \mathrm{span}\{\Phi_2(x_i) - \Phi_2(x_j) : 1 \le i < j \le n\}.$$

There exists a randomized algorithm that, given X, π, and $0 < \varepsilon, \delta < 1$, outputs a ranking $\widehat{\sigma}$ induced without ties by a quadratic score and runs in time

$$2^{O((r_2 \log(2/\varepsilon) + \log(2/\delta))/\varepsilon^2)}\,\mathrm{poly}(n, d, L_X).$$

With probability at least $1 - \delta$, $\widehat{\sigma}$ has Kendall agreement at least $1 - \varepsilon$ times the largest agreement of any quadratic score on X. It is a PTAS when $r_2 = O(\log n)$, in particular when $d = O(\sqrt{\log n})$. For $d = O(\log n)$ and fixed ε, δ, its running time is $2^{O(\log^2 n)}\,\mathrm{poly}(n, L_X)$.

Proof. Apply Theorem G.1 with $p = 2$. The monomials of degrees one and two give exactly the unrestricted quadratic family. The original dimension threshold becomes $d = O((\log n)^{1/2})$, and the logarithmic-dimension running time becomes $2^{O(\log^2 n)}\,\mathrm{poly}(n, L_X)$. These ambient-dimension conditions suffice for the stated bounds; the algorithm's actual dependence is on r_2, which can be smaller than D_2. □

G.5 Hardness for every fixed degree

Feature expansion alone does not extend the linear hardness results. A polynomial score can realize additional orders and may therefore achieve a smaller distance on an instance that is hard for linear scores. We construct item blocks that let us obtain weights for the original inequalities from *any* ranking induced by a polynomial score. The resulting approximation factor increases by at most a constant that depends on the fixed degree. We can absorb this constant by decreasing the exponents in Theorem D.1.

G.5.1 Extracting a linear coefficient. We add one coordinate, denoted by t, and choose p fixed *evaluation values* for it. A weighted sum of the scores at these values isolates the coefficient of t^{p-1}. Since the total degree is at most p, this coefficient consists of a constant and linear terms in the remaining coordinates. Evaluating at opposite feature vectors and taking the difference will cancel all but the inner product needed for the original inequality.

For $i \in \{0, \dots, p-1\}$, set $t_i = i - (p-1)/2$ and $c_i = (\prod_{j \ne i}(t_i - t_j))^{-1}$. The evaluation values are distinct rationals and sum to zero. When $p = 1$, the empty product equals 1, so $t_0 = 0$ and $c_0 = 1$. For a polynomial $f(\rho, x, t)$, where $\rho \in \mathbb{R}$ and $x \in \mathbb{R}^q$, define the evaluation sum $\mathcal{E}_p f(\rho, x) = \sum_{i=0}^{p-1} c_i f(\rho, x, t_i)$. Lemma G.5 shows that taking the difference of these sums at opposite feature vectors gives a linear inner product. We will use it to turn ranking comparisons into inequalities on the extracted coefficient vector.

Lemma G.5. *For every polynomial $f(\rho, x, t)$ of total degree at most p, there are coefficients $\beta_0, \beta_\rho \in \mathbb{R}$ and $b \in \mathbb{R}^q$ such that $\mathcal{E}_p f(\rho, x) = \beta_0 + \beta_\rho \rho + \langle b, x \rangle$. Consequently,*

$$\mathcal{E}_p f(\rho, a) - \mathcal{E}_p f(\rho, -a) = 2\langle b, a \rangle \qquad \text{for every } a \in \mathbb{R}^q.$$

These coefficients are rational when f has rational coefficients.

Proof. Let $L_i(t) = \prod_{j \ne i}(t - t_j)/(t_i - t_j)$. This polynomial equals 1 at t_i, equals 0 at every other evaluation value, and has leading coefficient c_i. For $0 \le k \le p-1$, the interpolation identity $t^k = \sum_i t_i^k L_i(t)$ holds: both sides have degree at most $p-1$ and agree at the p distinct evaluation values, so their difference must be zero. Comparing coefficients of t^{p-1} gives $\sum_i c_i t_i^k = 0$ for $k < p-1$, and gives 1 when $k = p-1$.

We must also eliminate the possible degree-p term. The polynomial $t^p - \prod_i (t - t_i)$ has degree at most $p-1$ and equals t_i^p at each evaluation value. Its coefficient of t^{p-1} is $\sum_i t_i = 0$, so the same interpolation argument gives $\sum_i c_i t_i^p = 0$. This argument also covers $p = 1$.

Collecting powers of t gives $f(\rho, x, t) = \sum_{k=0}^{p} f_k(\rho, x) t^k$. The identities just proved give $\mathcal{E}_p f(\rho, x) = f_{p-1}(\rho, x)$. The total-degree bound implies that f_{p-1} has degree at most one in (ρ, x), so it consists of a constant and linear terms. Evaluating at a and $-a$ cancels its constant and its ρ term. For rational f, its coefficients, including b, are rational as well. □

G.5.2 Encoding one inequality in an item block. Consider the homogeneous strict inequality system $\mathcal{I} = (\langle a_\ell, z\rangle > 0 : \ell \in [m])$ from Section 5.1, with nonzero rational $a_\ell \in \mathbb{Q}^q$. Let S be the number of bits needed to describe this system. As before, repeated inequalities count separately. Choose $v_i = i + 1$ when $c_i > 0$, and $v_i = -(i + 1)$ otherwise. Let g be the rational polynomial of degree at most $p - 1$ that satisfies $g(t_i) = v_i$; explicitly, $g(t) = \sum_i v_i L_i(t)$.

For each inequality, create the $2p$ items

$$u_{\ell,\eta,i} = (m - \ell, \eta a_\ell, t_i), \qquad \eta \in \{-1, 1\}, \quad 0 \le i < p.$$

Let π be the target ranking that lists blocks in increasing ℓ. Inside a block, it lists the items in decreasing order of ηv_i. The values v_i have distinct absolute values, so the $2p$ numbers ηv_i are distinct. The coefficient ηc_i has the same sign as ηv_i. Thus every item with a positive coefficient in the extraction sum precedes every item with a negative coefficient. We will show that any polynomial score that preserves this block order must give a positive inner product for the corresponding inequality.

Set $K_p = p(2p - 1)$ and $F_p = 2p^2$. These are the Kendall's Tau and footrule costs of reversing a block of $2p$ consecutive items. Lemma G.6 relates the ranking distances to the minimum violation-count. Its lower bounds also hold when the output mixes items from different blocks.

Lemma G.6. *For every fixed $p \ge 1$, given a homogeneous strict system $\mathcal{I} = (\langle a_\ell, z\rangle > 0 : \ell \in [m])$ in q variables, the construction above produces $n = 2pm$ pairwise distinct rational items in dimension $d = q + 2$ and satisfies*

$$\mathrm{OPT}_{\mathrm{viol}} \le \mathrm{OPT}_{\mathrm{KT}}^{(p)} \le K_p \,\mathrm{OPT}_{\mathrm{viol}},$$
$$2\,\mathrm{OPT}_{\mathrm{viol}} \le \mathrm{OPT}_{\mathrm{FR}}^{(p)} \le F_p \,\mathrm{OPT}_{\mathrm{viol}}\,.$$

For every fixed p, there exists a polynomial-time algorithm that, given any ranking σ induced without ties by a polynomial score of degree at most p, outputs a rational vector b with $\mathrm{viol}_{\mathcal{I}}(b) \le d_{\mathrm{KT}}(\pi, \sigma)$ and $2\,\mathrm{viol}_{\mathcal{I}}(b) \le d_{\mathrm{FR}}(\pi, \sigma)$. The construction also takes polynomial time for each fixed p, and the coordinates of the constructed items and the returned vector have polynomial-length rational representations.

Proof. *Distinctness and encoding.* Different blocks have different first coordinates. Within a block, the distinct evaluation values distinguish different i, and the two choices of η differ because $a_\ell \ne 0$. All items are therefore distinct, even when coefficient vectors repeat. The numbers t_i, c_i, v_i and the coefficients of g depend only on p. For fixed p, creating the items takes polynomial time in S and gives polynomial encoding length.

Constructing a ranking from inequality weights. Choose weights z that minimize the violation-count. The perturbation and rational feasibility argument in Appendix D.1 let us choose rational z with $\langle a_\ell, z\rangle \ne 0$ for every ℓ, without increasing its violation-count. Put $A = \max_{\ell,i} |v_i \langle a_\ell, z\rangle|$, choose $\lambda = 2A + 1$, and use the degree-at-most-p score

$$f(\rho, x, t) = \lambda\rho + g(t)\langle z, x\rangle.$$

At an item in block ℓ, this score is $\lambda(m - \ell) + \eta v_i \langle a_\ell, z\rangle$. For any $\ell < \ell'$, every item in block ℓ has score at least $\lambda - 2A = 1$ greater than every item in block ℓ', so the blocks remain consecutive and follow their target order. If the inequality holds, the order within its block agrees with the target. If it fails, its nonzero inner product is negative, and the order within the block is the reverse of its target order. All scores are distinct. Each failing inequality contributes $\binom{2p}{2} = K_p$ inversions and $\sum_{j=1}^{2p} |2p + 1 - 2j| = F_p$ units of footrule distance. Different blocks create no other errors. This proves the two upper bounds because z minimizes the violation-count.

Obtaining inequality weights from any realizable ranking. Expand the constructed items into their monomial vectors and solve the adjacent-score constraints for the returned order, requiring each score difference to be at least 1. A polynomial score that induces the ranking without ties makes this rational linear program feasible after positive scaling. Its rational solution has polynomial encoding length for fixed p, because the proof of Theorem G.1 bounds the encoding length of the expanded coordinates. Interpret its weights as a polynomial f and obtain b from Lemma G.5. For each inequality occurrence, define $H_\ell = \sum_i c_i [f(m - \ell, a_\ell, t_i) - f(m - \ell, -a_\ell, t_i)]$. Lemma G.5 gives $H_\ell = 2\langle b, a_\ell\rangle$.

Suppose the returned ranking keeps the target order within block ℓ. Every item with positive coefficient ηc_i then has greater score than every item with negative coefficient. Let m_+ be the smallest score in the positive-coefficient group, and let m_- be the largest score in the negative-coefficient group. We have $m_+ > m_-$. The sum of the positive coefficients is $W = \sum_i |c_i| > 0$, and the negative coefficients sum to $-W$. Consequently, $H_\ell \ge W(m_+ - m_-) > 0$. If $\langle b, a_\ell\rangle \le 0$, the output must therefore reverse at least one pair within that block. These pairs belong to distinct blocks, so the violation-count is at most the Kendall's Tau distance.

For footrule, choose one reversed pair from each such block. Suppose its two items occupy target positions $j < k$, while those same items occupy output positions $j' > k'$, respectively. Their contribution is at least $|j' - j| + |k' - k| \ge (j' - k') + (k - j) \ge 2$. Different blocks have disjoint items, so summing these bounds gives twice the violation-count as a lower bound on footrule distance. This argument does not require output blocks to remain consecutive. No resulting weight vector can have fewer violations than $\mathrm{OPT}_{\mathrm{viol}}$, which proves both optimum lower bounds. Solving the rational linear program and extracting the coefficient vector also proves that this conversion takes polynomial time. □

For Kendall's Tau, an α-approximation on this instance gives weights with at most $K_p \alpha\,\mathrm{OPT}_{\mathrm{viol}}$ violations. For footrule, it gives at most $p^2 \alpha\,\mathrm{OPT}_{\mathrm{viol}}$ violations, because Lemma G.6 bounds their number by half the returned footrule distance. Both factors are at most $C_p \alpha$, where $C_p = K_p$. Since the degree is fixed, C_p does not grow with the number of inequalities or the feature dimension. The lower bounds below use this fact to absorb the additional factor.

G.5.3 Approximation hardness and dimension dependence. Theorem G.7 now applies the inequality lower bounds used for linear

scores. The additional factor C_p changes only their exponent constants. The result allows an algorithm to use every coefficient of degree at most p, rather than just the special polynomials constructed in the preceding proof.

Theorem G.7. *For every fixed integer $p \geq 1$, consider instances consisting of $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π of X, where d is part of the input. There are constants $c_p, c'_p > 0$ such that the polynomial minimum-disagreement problems under Kendall's Tau and Spearman footrule distances admit no deterministic polynomial-time $n^{c_p/\log\log n}$-approximation unless* P = NP. *Under PGC, a deterministic polynomial-time $n^{c'_p}$-approximation for either objective would also imply* P = NP.

Proof. Apply Lemma G.6 to the inequality systems in Theorem D.1, whose lower bounds follow from [60]. Let $c_0, c_1 > 0$ be their exponent constants. A ranking approximation of factor α would give an approximation of factor at most $C_p\alpha$ for the violation-count. Both conversions take polynomial time for fixed p, and $n = 2pm \leq 2pS$.

For large S, we have $2pS \leq S^2$. The ratio $\log t/\log\log t$ eventually increases with t, so

$$\log\left(C_p n^{c_p/\log\log n}\right) \leq \log C_p + \frac{2c_p \log S}{\log\log S}.$$

Choose $0 < c_p < c_0/4$. The right side is strictly less than $c_0 \log S/\log\log S$ for sufficiently large S. Thus the resulting approximation factor is smaller than the lower bound $S^{c_0/\log\log S}$, a contradiction. For the PGC bound, choose $0 < c'_p < c_1/2$. Then $C_p(2pS)^{c'_p} < S^{c_1}$ for sufficiently large S, giving the other contradiction. As in the main hardness statements, the factor expressions are asymptotic; changing their values for the finitely many smaller item counts does not affect either contradiction. □

Removing redundant original coordinates must preserve the entire polynomial scoring family, not just its linear part. Substituting linear expressions plus constants preserves the maximum degree and gives the required correspondence. Theorem G.8 expresses hardness both in the supplied feature dimension and in the effective dimension of the expanded features, which controls the algorithms for polynomial scores.

Theorem G.8. *For every fixed integer $p \geq 1$, consider instances consisting of $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π of X. Let r be the dimension of the span of the vectors $x_i - x_j$, and let r_p be the dimension of the span of the vectors $\Phi_p(x_i) - \Phi_p(x_j)$, in both cases over all $1 \leq i < j \leq n$. There are constants $a_p, b_p > 0$ such that the polynomial minimum-disagreement problems under Kendall's Tau and Spearman footrule distances admit no deterministic polynomial-time $d^{a_p/\log\log d}$-approximation unless* P = NP. *Under PGC, a deterministic polynomial-time d^{b_p}-approximation for either objective would also imply* P = NP. *The same conclusions hold with r_p in place of d, after decreasing the positive constants if necessary. All these bounds hold even when the original ambient and effective dimensions coincide, $d = r$. We interpret the first factors asymptotically, with any fixed bound in the finitely many smaller dimensions.*

Proof. Start with the block instance in $\mathbb{Q}^{q+2}$. Let $u_j = u_1 + Py_j$ express its items in a rational basis, where the matrix P has full column rank r and the differences of the y_j span $\mathbb{R}^r$. Gaussian elimination computes these coordinates in polynomial time, as in the preliminaries. The rational matrix $T = (P^\top P)^{-1}P^\top$ satisfies $TP = I_r$.

Every original polynomial f gives a polynomial $h(y) = f(u_1 + Py)$ of degree at most p on the compressed coordinates. Conversely, every such h extends to $f(u) = h(T(u - u_1))$, also of degree at most p, and $f(u_j) = h(y_j)$ at every input item. These substitutions show that both polynomial families induce exactly the same rankings without ties and have the same optimum values. They also preserve distinctness: $u_i = u_j$ if and only if $y_i = y_j$, because P has full column rank. The compressed coordinates have polynomial encoding length. Given any returned compressed ranking, solve its adjacent-score constraints in the compressed monomial features to obtain a rational polynomial h. Composing h with $T(u - u_1)$ gives a rational polynomial that induces the same ranking on the original block items. We then obtain weights for the original inequalities using Lemma G.6. These operations have polynomial encoding length and running time for fixed p.

Rename the compressed ambient dimension as d; then $d = r$ and $d \leq q + 2 \leq S + 2$. Its expanded effective dimension satisfies $r_p \leq \binom{d+p}{p} - 1 = O_p(d^p)$. For either dimension measure $\xi \in \{d, r_p\}$, we therefore have $\xi \leq S^{2p}$ for sufficiently large S. For large ξ, eventual monotonicity gives

$$\log\left(C_p \xi^{a_p/\log\log \xi}\right) \leq \log C_p + \frac{2pa_p \log S}{\log\log S}.$$

Choose $0 < a_p < c_0/(4p)$ to make this strictly smaller than $c_0 \log S/\log\log S$ for sufficiently large S. For PGC, choosing $0 < b_p < c_1/(4p)$ gives $C_p \xi^{b_p} \leq C_p S^{2pb_p} < S^{c_1}$. Either such dimension-dependent ranking algorithm would thus give an approximation for the violation-count ruled out by Theorem D.1. The factor expressions are again asymptotic, so the finitely many dimensions outside the monotonic range do not affect the conclusion. This proves the claims for both dimension measures on instances with $d = r$. □

The approximation lower bounds also exclude exact optimization for arbitrary dimension. Complementing Kendall's Tau disagreement gives the maximum-agreement consequence, and integrality rules out an FPTAS for each objective. These consequences do not require PGC.

Corollary G.9. *For every fixed integer $p \geq 1$, consider instances consisting of $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π of X, where d is part of the input. Unless* P = NP, *no deterministic polynomial-time exact algorithm or deterministic FPTAS exists for minimizing Kendall's Tau distance, minimizing Spearman footrule distance, or maximizing Kendall agreement.*

Proof. An exact algorithm for either minimum distance would contradict Theorem G.7. For every strict ranking, Kendall agreement equals $\binom{n}{2}$ minus Kendall's Tau distance. Optimizing agreement over the same polynomial ranking family would therefore optimize Kendall's Tau distance exactly, giving the same contradiction.

Now suppose one of these objectives admits an FPTAS. Run it with $\varepsilon = 1/(2n^2)$. Every feasible objective is an integer smaller than n^2. For either minimum distance, the returned value exceeds

optimum by less than 1/2; for agreement, the optimum minus the returned value is less than 1/2. Its integer value must consequently equal the optimum. The FPTAS runs in time polynomial in the binary input size and $1/\varepsilon = 2n^2$, producing the excluded exact algorithm. □

Scope of the logarithmic-dimension claim. For $p \geq 2$, Lemma G.6 controls approximation gaps only up to a degree-dependent constant. Its bounds do not preserve the one-unit clique gap in Theorem D.2 in Appendix D.3. The reduction therefore does not establish exact or FPTAS hardness for higher-degree scores in logarithmic dimension. For $p = 1$, the linear theorem applies directly. For higher degrees, the hardness bounds concern arbitrary dimension, whereas the agreement PTAS applies when $r_p = O(\log n)$.

H MISSING DETAILS FROM SECTION 6: DATASETS AND ADDITIONAL RESULTS

H.1 Dataset description

- **NBA**: Following prior work on related problems [15], we use the NBA dataset [80], which contains 22,840 tuples describing player statistics across the 1979/80–2022/23 seasons. Each tuple corresponds to a player-season combination, uniquely identified by the PLR attribute, comprising the player's name, age, and team. The default ranking attributes are the player's season-average statistics: PTS, REB, AST, STL, BLK, FG%, 3P%, and FT%. In line with the previous work, to eliminate duplicate rankings, we retain only one player-season record when multiple records have identical values across these attributes. We consider two sources of ground-truth rankings: the MVP ranking and a ranking based on the product of Minutes Played (MP) and Player Efficiency Rating (PER). Computing PER involves a complex formula and requires additional attributes beyond those included in our default set of ranking attributes.
- **Retiring Adult** [25]: The Retiring Adult dataset consists of 1664500 records pertaining to the Income of individuals surveyed in the United States. We use a cleaned and numerically encoded version of the full Adult income dataset. Each record represents an individual and is described by ten attributes derived from demographic, educational, employment, and financial information. Ordinal (and categorical) attributes like Education are encoded numerically to facilitate algorithmic processing. The records are ordered according to the income attribute in non-increasing order, and the Income field is removed from the dataset. The ordering provides us with a ranking in need of an explanation.
- **IIT JEE 2009**: JEE is a yearly entrance exam, the scores of which are used as a criterion for admission to the top engineering colleges in India. The dataset consists of Physics, Mathematics, and Chemistry scores for 384,977 candidates from the year 2009. The dataset also captures demographic information along with the scores themselves. As the ranking is not available for this dataset, we follow two models for producing a viable ranking — the Additive bonus and Multiplicative bonus models from Yan et al. [89]. The dataset has 2 minority groups, SC and ST. We use unit weights for Physics, Mathematics, and Chemistry, but account for "*historical biases*" through an additive/multiplicative bonus. For the additive and multiplicative bonuses, we set the bonus such that the difference between the mean of each minority group and the mean of the majority group is minimized, along with ℓ_2-regularization to prevent arbitrarily large bonus scores. Note that determining the additive/multiplicative bonus is a convex optimization problem that can be efficiently solved. Thus, we generate two datasets with rankings based on the additive bonus model and the multiplicative bonus model. The final dataset only consists of the Physics, Chemistry and Mathematics marks of the candidates.
- **Human Cognitive Performance**: We use the Human Cognitive Performance Analysis dataset [73], obtained from Kaggle. Each tuple represents an individual and records demographic characteristics, lifestyle factors, and measures related to cognitive performance. The dataset contains a reported 80,000 records and 11 attributes: age, gender, sleep duration, stress level, diet type, screen time, exercise frequency, caffeine intake, reaction time, cognitive score, and memory test score. We retain 8 numerical attributes (age, sleep duration, stress level, screen time, exercise frequency, caffeine intake, reaction time, and memory test score) and sort the dataset based on cognitive score in non-increasing order.
- **Synthetic dataset**: The synthetic datasets consist of a collection of 7 datasets. The data records within each dataset follow a uniform distribution within a sphere of radius 100. As the scores in this dataset are not available, we pick a linear scoring function to generate a ranked order and then perturb it using the Mallows model [26, 53], which is defined as follows:

$$P(\sigma \mid \sigma_0, \phi) = \frac{\phi^{d_{\mathrm{KT}}(\sigma,\sigma_0)}}{Z_n(\phi)}$$

where $Z_n(\phi) = \prod_{j=1}^{n} \left(1 + \phi + \ldots + \phi^{j-1}\right)$ and ϕ is called the dispersion parameter. In our experiments, we choose ϕ to correspond to an expected number of inversions. We generate 2 kinds of datasets: (i) varying dimensions (3, 7, 11, 15) when the number of tuples is fixed at 1,000,000 (with ϕ chosen such that the selected permutation has 5% inversions), and (ii) varying dataset size (100,000, 200,000, 400,000, 1,000,000) when the number of dimensions is fixed at 3 (with ϕ chosen such that the selected permutation has 5% inversions in expectation). Thus, overall we generate 7 datasets as combinations of parameters.

H.2 Additional experimental results, tables and figures

H.2.1 Effects of unknown scoring models. In the HCP dataset (Table 10), `MaxFS` and `Sampling` (weighted and unweighted) perform nearly as well as each other, closely followed by the `RankSVM` and `RankNet` methods. `Sampling-CLR` and `LMRC` follow these methods but with a noticeable increase in mean-KT scores. The standard deviations for the different methods show that weighted sampling is the most consistent after `LMRC` (deterministic), followed by the sampling algorithm. `MaxFS` follows closely next. `Approximate-Agreement` does not perform as well due to the smaller values of ϵ that can be targeted within the 30-minute time limit.

Table 9 shows the results for the NBA dataset. Due to the smaller size of the NBA dataset, many approaches are able to complete within the 30-minute timeout. `LS` is able to complete within the timeout and performs the best among the different methods. It is

Table 11: Comparison to OPT for small uniform dataset

Method	Kendall-Tau
OPT Local Search, Sampling, MaxFS	286
Weighted sampling	290
Sampling-CLR, LMRC, Approx-Agree	291
RankNet, RankSVM	292
CLR	293
1/2-approx	2143

Table 9: Per-method Kendall's Tau statistics on the NBA dataset. Lower values are better. Values are reported over 5 different runs with different seeds.

Method	Mean	Median	Std. Dev.
LocalSearch	13206428.4	13205293	1827.9
RankSVM	13214827.4	13215616	2702.1
RankNet	13221703.2	13223065	5292.1
Weighted Sampling	13226311.6	13227459	7520.6
Sampling-CLR	13241848	13243073	2996.1
MaxFS	13245519.4	13237678	23148.7
Sampling	13256424.8	13254931	9961.0
LMRC	14025325	14025325	0
Approx-Agree	18647509.2	16016564	4619031.3
1/2-approx	76758870.4	78414135	31138648.0077

Table 10: Per-method Kendall's Tau statistics on the Human Cognitive Performance dataset. Lower values are better. Values are reported over 5 different runs with different seeds.

Method	Mean	Median	Std. Dev.
MaxFS	240441393	240324386	307709.8
Sampling	240460696.4	240424216	263291.4
Weighted Sampling	240579932.8	240588720	116457.7
RankSVM	242121991.4	242061497	832360.5
RankNet	242443678.2	242301239	1193067.4
Sampling-CLR	249681766.8	248843834	2063174.2
LMRC	250517167	250517167	0
Approx-Agree	314840422	306580969	27043109.5
1/2-approx	609923813.8	613372739	17203349.4

closely followed by RankSVM, RankNet, and Weighted Sampling. Sampling-CLR and MaxFS compete closely except for the large deviations shown by the MaxFS approach. Deterministic LMRC shows no deviations but does not perform as well as other methods.

H.3 Explanation using linear models — including OPT

We start our experiments with the class of linear functions and run all algorithms on the synthetic dataset with 3 dimensions. The dataset is restricted to 100 rows which helps us efficiently compute OPT using an ILP formulation. The results are presented in Table 11. With $n = 100$ and optimum disagreement (KT score) 286, the optimum agreement is $4950 - 286 = 4664$. CLR's disagreement ratio is $293/286 = 1.0245$; Approx-Agree's agreement ratio is $4659/4664 = 0.9989$, which exceeds the $1 - \epsilon = 0.75$ target. The half-approximation's agreement ratio is $(4950 - 2143)/4664 = 0.6018$, which is greater than $1/2$. Thus, the results corroborate our approximation factor of $d + 1$ for Closest-Linear Ranking and better than $1/2$ for 1/2-approx. Approx-Agree was experimented with $\epsilon = 0.25$ and the number of Agreements is 4659 which gives a better ratio than $1 - \epsilon = 0.75$.

H.4 Preliminary experiments for degree 2 polynomial

We perform preliminary experiments with degree-2 polynomials. For this experiment, we use the Uniform dataset with 3 dimensions and $1,000,000$ data points. The results of this experiment are presented in Table 12. RankRLS performs the best in mean value, closely followed by Weighted Sampling, Sampling and LMRC, all three of which have the same KT value. Note that this occurs due to a similar optimization function being at the core of all three approaches. SmoothedMRC, Sampling-CLR and MaxFS show similar results for KT value. RankSVM performs better than RankNet, however, both approaches perform the worst among all approaches. Note that these are preliminary experiments on a single dataset and do not provide conclusive evidence about the behavior of all approaches. We plan to perform extensive experiments to validate our methods more thoroughly.

Table 12: Per-method Kendall's Tau statistics on the Uniform dataset with 3 dimensions and 1,000,000 data points for degree 2 ranking functions. Lower values are better.

Method	Mean	Median	Std. Dev.
RankRLS	249426503.0000	249426503.0000	0.0000
Weighted Sampling	249426665.0000	249426665.0000	0.0000
Sampling	249426665.0000	249426665.0000	0.0000
LMRC	249426665.0000	249426665.0000	0.0000
Smoothed MRC	249471591.2000	249459045.0000	29161.1090
Sampling-CLR	249765666.2000	249745632.0000	106915.1907
MaxFS	250126690.4000	250169615.0000	309577.4485
RankSVM	260494195.6000	260793267.0000	969434.8137
RankNet	282744668.6000	282910891.0000	705554.1979

I MISSING DETAILS FROM SECTION 6: POLYNOMIAL EXPERIMENTS

Experimental setup. We extend the comparison in Section 6 to polynomial scoring functions using the representation in Appendix G. For a maximum total degree p, each item x_i is represented by the vector $\Phi_p(x_i)$ of all nonconstant monomials of degrees $1, \ldots, p$. We seek a score $f_a(x_i) = a^\top \Phi_p(x_i)$, with unrestricted real coefficients. The expanded dimension is $D_p = \binom{d+p}{p} - 1$, and $p = 1$ recovers the linear scoring family. The implementation defaults to quadratic scores, but supports any fixed degree subject to memory limits. All methods in a comparison use the same degree, input scaling, and monomial representation. Pairwise losses use $v_{ij} = \Phi_p(x_i) - \Phi_p(x_j)$, with $i \succ_\pi j$, and margin $m_{ij} = a^\top v_{ij}$.

Baselines. We compare against seven baselines restricted to this same polynomial scoring family. *Polynomial Linearized MRC* applies the ridge-regularized, linearized rank-correlation estimator of Shen et al. [75] to the expanded features. *Polynomial RankSVM* minimizes the regularized pairwise hinge loss $\max\{0, 1 - m_{ij}\}$ [42]; polynomial-kernel ranking models are also considered in prior work [71]. *Polynomial RankNet* instead minimizes the regularized pairwise logistic loss $\log(1 + \exp(-m_{ij}))$ [9], with the score restricted to f_a, rather than a neural network. *Polynomial RankRLS*

uses the preference-graph least-squares formulation of Pahikkala et al. [63]: it minimizes the mean squared preference error $(1-m_{ij})^2/2$, together with an ℓ_2 penalty. Each preference has unit target margin, so the method requires only the target order, not numerical target-score differences. Its matrix-free conjugate-gradient implementation avoids constructing the full pair matrix. *Smoothed MRC* maximizes the mean of $s(m_{ij}/h)$, where $s(t) = 1/(1+\exp(-t))$, over a unit coefficient direction. It uses multiple starts, projected ascent, and decreasing bandwidth h. This is a logistic-smoothing adaptation motivated by smoothed rank-correlation estimation [90], not a reproduction of its self-induced smoothing procedure. Unlike RankNet, it maximizes sigmoids rather than log-sigmoids. *Polynomial Kendall local search* applies the full-data coordinate and Gaussian search from Section 6 to polynomial coefficients [57]. Finally, *Polynomial MaxFS* applies elastic comparison-slack minimization, greedy deletion, and add-back to the lifted constraints [16]. The pairwise learners use all eligible comparisons when practical, or a bounded uniform sample of original item pairs. RankRLS also supports the full preference graph implicitly, while full-data RankSVM computes hinge subgradients without enumerating pairs. Every baseline returns a polynomial score; agreement and disagreement are recomputed on the full comparison population.

Details of our algorithms. The polynomial feature representation lets us reuse the comparison algorithms from the linear setting. We distinguish the full-instance minimum-disagreement approximation from the two sampled local-search variants used for scalable maximum agreement.

`Closest Polynomial Ranking`. We run `Closest Linear Ranking` on all eligible lifted comparisons. The algorithm repeatedly removes an inclusion-minimal infeasible subsystem until the retained comparisons admit a common coefficient vector, then refines its score order while preserving those comparisons. Theorem G.2 gives an (r_p+1)-approximation, where r_p is the dimension spanned by the lifted feature differences. Thus the factor depends on the expanded comparison space, with $r_p \le D_p$, not simply on the original number of attributes d. This method uses the full comparison population and is intended for instances on which the LP and infeasible-subsystem computations are practical.

`Sampling`. We sample eligible original item pairs uniformly with replacement and run the circle-sweep local search from Section 6 on their lifted differences. Repeated comparisons retain their sampling multiplicities. A thin singular value decomposition restricts the search to the span of the sampled differences, whose dimension is at most the number of distinct sampled comparisons. Starting directions include the polynomial Linearized MRC pilot, the weighted sample centroid, their reverses, and random directions. Each sweep searches the arcs between sampled comparison sign changes on a great circle and accepts only a numerically verified non-worsening move. Finalists and their negatives are evaluated on the full dataset.

`Weighted Sampling`. To reduce repeated sampling of comparisons between large duplicate classes, we group identical feature rows and use a mixture of the original item-level distribution, a class-balanced distribution, and a distribution favoring small-margin comparisons under the polynomial Linearized MRC pilot. The default mixture weights are 0.5, 0.25, and 0.25, respectively. For a directed class comparison e, let W_e count its eligible original item pairs and let $P = \sum_e W_e$. Its importance weight is proportional to $(W_e/P)/q(e)$, where $q(e)$ is its proposal probability. Thus the weighted search targets original-item agreement rather than giving each class comparison equal weight. Small instances use an explicit class-pair table; larger instances compute multiplicities only for proposed class pairs, avoiding quadratic storage in the number of distinct rows. We then use the same sampled circle-sweep search and full-data finalist selection as `Sampling`.

Both sampled variants are heuristics, not implementations of the exact empirical optimizer required by the polynomial agreement scheme. A fixed sample budget controls the search problem; ε and δ provide reference sample-size diagnostics rather than a PTAS guarantee. The default search uses eight restarts and at most 250 iterations per restart. Comparing each finalist with its reverse nevertheless retains the $1/2$ maximum-agreement guarantee: the two orientations partition the optimizable target comparisons, while target-strict duplicate pairs are unavoidable disagreements in either orientation.

Duplicates, ties, and evaluation. We preserve original-item multiplicities throughout training and evaluation. Target-tied pairs are excluded, identical input rows remain tied, and a target-strict pair of identical rows contributes one forced disagreement. Including degree-one monomials ensures that polynomial lifting does not merge distinct input rows merely because their higher powers coincide. Numerical score ties between distinct lifted point classes are refined and the resulting order is checked before evaluation. We report full-data KT disagreement, agreement, and runtime including feature construction. Sampling reduces the comparison-search cost, but the implementation still constructs the $n \times D_p$ feature matrix for the pilot and final evaluation. Products and optimization use double precision; numerical certificates are checked on the computed lifted features, rather than on exact symbolic polynomial evaluations.

J RELATED WORK ON DISTANCE FUNCTIONS

An important aspect of our work looks at similarity between rankings. Numerous functions like Spearman's footrule [78, 79], Kendall's Tau distance [45, 46] (also often called *Kemeny distance* [44]), Hamming distance on ranks [37], Ulam distance [83] have been studied in different applications like rank aggregation [1, 28, 87], information retrieval and web search, recommender systems, learning to rank, database query optimization, bioinformatics and gene-expression analysis, feature selection, and the evaluation of ranking stability in machine-learning models. In our context of university ranking, sports, etc. Kendall's Tau and Spearman's footrule are most appropriate. In top-k settings where top ranks matter over others, weighted notions like NDCG [40, 41] are applicable. However, for the applications we consider, rank itself does not play a major role; instead, the full ranked order matters.